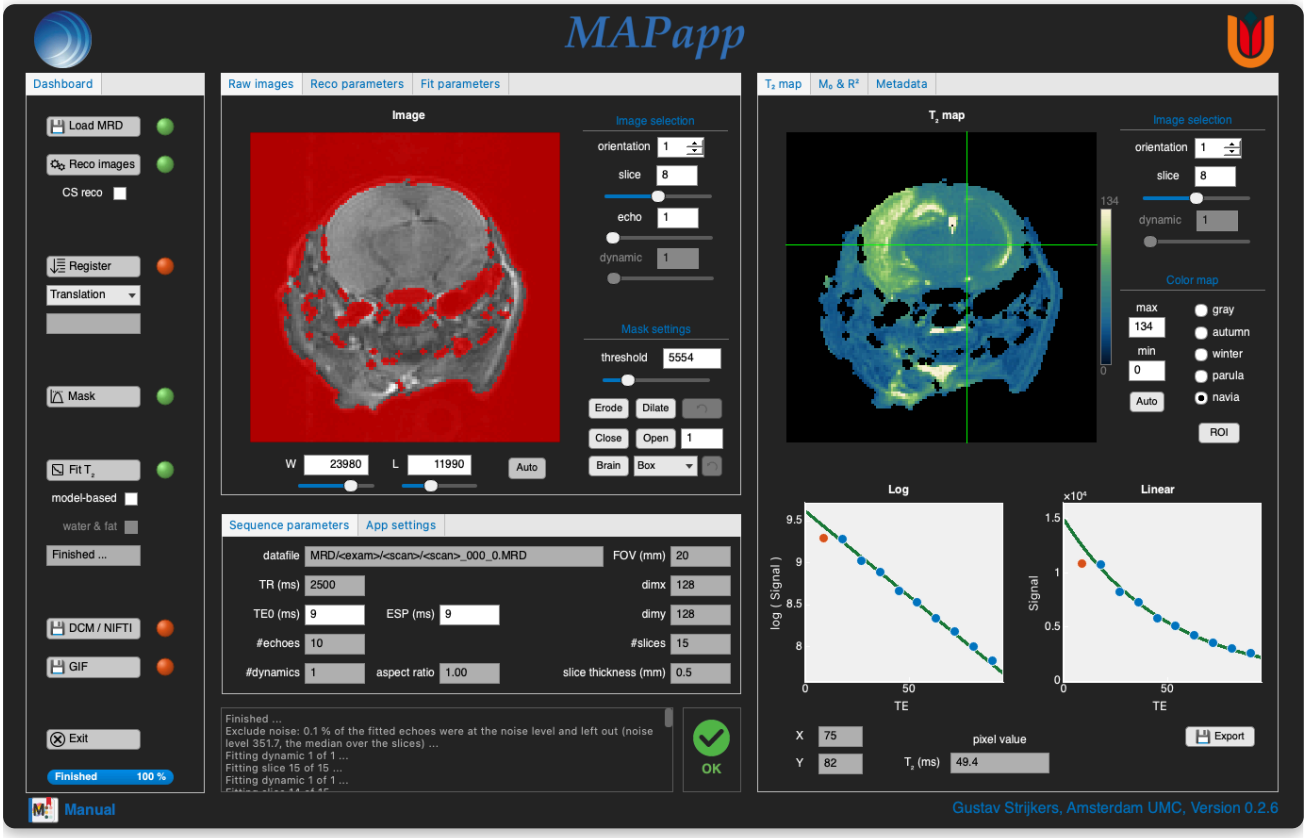
MAPapp — user manual

Version 0.3.3 (beta), November 2026



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1. Overview

MAPapp is an application with a graphical user interface to reconstruct relaxation, saturation transfer and diffusion maps from scans acquired with an MR Solutions preclinical MRI system. One window handles five kinds of scan:

- **T_1 scans**: inversion recovery, saturation recovery (a series of TRs) and variable flip angle series, fitted to T_1, M_0 and R^2 maps;
- **T_2 and T_2* scans**: multi-echo spin echo (T_2) and multi-gradient echo (T_2*), fitted to T_2 or T_2*, M_0 and R^2 maps; multi-gradient echo scans can also be separated into water and fat maps, with fat fraction and R_2* maps, and 3D multi-gradient echo scans give a magnetic susceptibility map (QSM);
- **ORACLE scans**: phase-cycled balanced steady-state free precession (bSSFP), which give T_1, T_2, M_0, off-resonance and R^2 maps from one scan;
- **CEST and MT scans**: a series of saturation offsets, from a few offsets to a full Z-spectrum, which give MTR, asymmetry (APT and other contrasts), fitted pool, B_0 and quality maps (section 9);
- **Diffusion scans**: diffusion-weighted EPI scans of two or more b-values, fitted to apparent diffusion coefficient (ADC), M_0 and R^2 maps (section 10).

The app recognizes the kind of scan when it is loaded and shows the map tabs and the Fit button that belong to it. The images are reconstructed, optionally registered, masked and fitted; the maps can be exported as DICOM, NIfTI, BIDS and GIF, with the values of a region of interest as a CSV file.

MAPapp is a stand-alone application: everything it uses, elastix and dcm2niix included, is inside it, and nothing else has to be installed.

2. Contact

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3. Disclaimer

This is a beta version. The software has been tested using mouse and rat data and phantoms acquired with an MR Solutions preclinical 7.0T system. However, this does not warrant that the functions contained in the program will meet your requirements or that the operation of the program will be uninterrupted or error-free. In case of questions or issues, please contact Gustav Strijkers.

4. Installation

macOS

Requirements: a Mac with Apple silicon (M1 or later), macOS 26 or later.

MAPapp comes as a disk image. Nothing else has to be installed: Python, elastix, dcm2niix and everything else the app uses are inside it.

1. Open the disk image and drag **MAPapp** onto the Applications folder in its window.
2. Open it from Applications. The app is not signed with an Apple Developer ID, so the first time macOS refuses it ("Apple could not verify ..."); click **Done**. Then open **System Settings > Privacy & Security**, scroll down to the line saying MAPapp was blocked, click **Open Anyway** and confirm. From then on it opens as any other app.

   Instead of step 2, the same in Terminal, once:

   ```
   xattr -dr com.apple.quarantine /Applications/MAPapp.app
   ```

The first start can take a little longer: macOS checks the app once.

Windows

Requirements: a PC with Windows 10 or 11, 64-bit.

MAPapp comes as an installer. Nothing else has to be installed.

1. Run the installer. It is not signed, so the first time Windows may stop it ("Windows protected your PC"): click **More info** and then **Run anyway**.
2. The installer puts the app in your user folder, without administrator rights (installing it for all users is an option), with a Start-menu entry and, if you choose, a desktop shortcut.

To remove the app, use **Settings > Apps**.

BART (optional)

With BART, the Berkeley Advanced Reconstruction Toolbox (mrirecon.github.io/bart), version 1.0.00 or newer, the model-based fits are available. Without BART, or with an older version, the model-based fit is switched off and compressed sensing runs on the built-in sigpy engine. The message box says so at start.

Installing BART on macOS with MacPorts

1. Install the Xcode command line tools:

   ```
   xcode-select --install
   ```

2. Install MacPorts from www.macports.org.
3. Install the packages:

   ```
   sudo port install fftw-3-single gcc14 libpng openblas flock gmake coreutils l
   ibblocksruntime
   ```

4. Download and unpack BART:

```
wget https://github.com/mrirecon/bart/archive/v1.0.00.tar.gz
tar xvfz v1.0.00.tar.gz
cd bart-1.0.00
```

5. Build and install:

```
export PATH=/opt/local/libexec/gnubin:$PATH
CC=gcc-mp-14 gmake
sudo gmake install
```

   The compiler name must match the gcc version installed in step 3: gcc14 gives gcc-mp-14, gcc15 gives gcc-mp-15.

6. Check that it works: `bart version`

Installing BART on macOS with Homebrew

1. Install the Xcode command line tools: `xcode-select --install`
2. Install Homebrew:

```
/bin/bash -c "$(curl -fsSL https://raw.githubusercontent.com/Homebrew/instal
l/HEAD/install.sh)"
```

3. Install the packages BART needs:

```
brew install gcc libpng fftw openblas make llvm libomp
```

4. Download and unpack BART as above (step 4 of MacPorts).
5. Build and install:

```
CC=gcc-15 MACPORTS=0 gmake
sudo gmake install
```

   The compiler name must match the version Homebrew installed; check it with `ls /opt/homebrew/bin/gcc-*`.

6. Check that it works: `bart version`

How MAPapp finds BART on macOS

`sudo make install` puts BART in `/usr/local/bin`, which is where the app looks first. It looks in these folders, in this order:

Folder	
`/usr/local/bin`	where BART's own `make install` puts it
`/usr/bin`	
`/opt/homebrew/bin`	Homebrew's
`/opt/local/bin`	MacPorts'

An app opened from the Finder or the Dock does not see the settings of your shell (such as `TOOLBOX_PATH` or `PATH` in your shell profile). If your `bart` is in another folder, a link to it in `/usr/local/bin` lets the app find it. In Terminal, once (Terminal asks for your password):

```
sudo mkdir -p /usr/local/bin
sudo ln -s /path/to/bart/bart /usr/local/bin/bart
```

Installing BART on Windows 11

On Windows, BART runs inside the Windows Subsystem for Linux (WSL); the app calls it as `wsl bart`.

1. Install WSL. Start a Windows PowerShell as administrator and run `wsl --install`. This installs WSL 2 and Ubuntu; a restart is needed. The first time you start the Linux prompt you are asked to create a user account.
2. Update the Linux distribution:

   ```
   sudo apt-get update
   sudo apt-get dist-upgrade
   ```

3. Install the packages BART needs (gcc 12 or newer):

   ```
   sudo apt-get install make gcc libfftw3-dev liblapacke-dev libpng-dev libopenb
   las-dev gfortran
   ```

4. Download, build and install BART:

   ```
   wget https://github.com/mrirecon/bart/archive/v1.0.00.tar.gz
   tar xvfz v1.0.00.tar.gz
   cd bart-1.0.00
   make
   sudo make install
   ```

5. Check from a Windows command prompt: `wsl bart version`

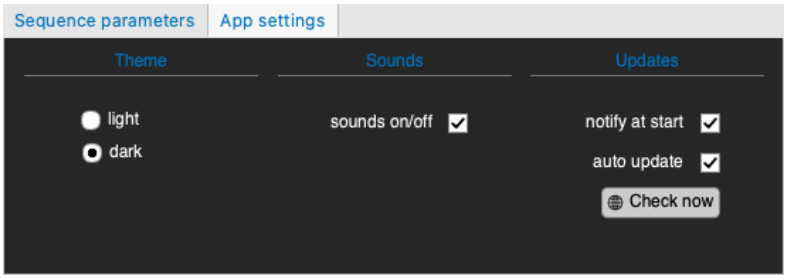
5. Running the software

Open **MAPapp** from the Applications folder (macOS), or from the Start menu or the desktop shortcut (Windows). The window appears within seconds. At its start the app looks for elastix and BART, and the message box reports what it found; "Ready" says it is done. Of the steps, only **Load MRD** and **Exit** are on until a scan is loaded (and, in the Reco parameters tab, **gather single-offset scans**, which the load reads).

A scan given on the command line (an MRD file or its folder, as a program that starts MAPapp with a scan gives it) is loaded at once, as with **Load MRD**; a path the app cannot load gives a warning in the message box. When the app is open already, a scan given this way goes to the open window instead: the window comes to the front and loads it, or loads it when the step under way ends.

Updates, in the App settings tab (beside the Sequence parameters tab):

- **auto update**, ticked as it comes: new versions of MAPapp arrive by themselves. When one is ready, a window says so with what is new: **Install and Restart** installs it and starts the new MAPapp; **Later** installs it when you quit MAPapp, without asking again. The window does not come while a step runs, but when it has finished. Without an internet connection nothing is said and nothing waits.
- **Check now** looks for a newer version at once, also with **auto update** off; the message box says what it found. On the Mac, **Check for Updates...** in the app menu does the same.
- **notify at start**, ticked as it comes, counts only with **auto update** off: when a newer version is available, a window at the start offers it. **Download** gets the installer for this computer, **Skip this version** shows no window for that version again, **Later** asks again at the next start.



The App settings tab: theme, sounds and updates

Theme and sounds. The App settings tab, with its Theme, Sounds and Updates panels side by side, also chooses the window's theme, **light** or **dark**, and switches the sounds that go with a warning or an error (**sounds on/off**).

This manual. The **Manual** button at the bottom of the window opens this manual in the computer's PDF viewer.

The window has a fixed size. On a screen too small for it, the window fits the screen and can be scrolled.

Settings and log files

MAPapp keeps its files in a folder of its own, one per version, which it makes at its first start:

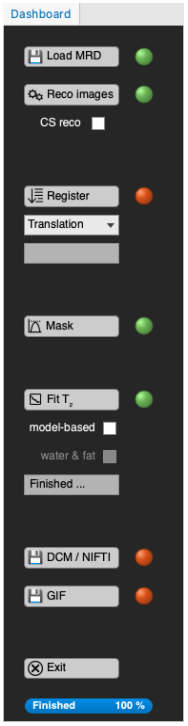
macOS	`~/Library/Application Support/MAPapp/<version>p`
Windows	`%APPDATA%\MAPapp\<version>p`

Settings. The settings are kept in `mapappSettings.json`. The file is written with the standard values the first time a version starts, with your choices from the version used before, and can then be edited by hand; changes take effect at the next start. It holds the version check, the folders where loading and the exports start (`mrdImportPath`, `defaultExportPath`), the theme, the sounds, the values the reconstruction, registration and fit fields start with, the color maps of the maps, and the number of log files kept. A value that cannot be used is replaced by its standard value, and the message box says so; a file that cannot be read at all is set aside as `mapappSettings-unreadable-<time>.json` and replaced by a new one.

Kept choices. MAPapp keeps these choices for the next start, and a new version takes them over: the theme, the sounds, auto update and notify at start, the color maps of the map tabs (T_1, T_2, M_0, R^2, water, fat, off-resonance, susceptibility, fat fraction, R_2^*, and the four CEST/MT tabs), the registration method, min R-square, de-noise, de-ring, exclude noise, the ORACLE smoothing σ, the methods and parameters of the water & fat and QSM method tabs, the CEST/MT options (not the contrasts' ticks or the linked maps), the folders last used for loading and for the exports, and the lab values of the Metadata tab (step 9).

Log files. Every session writes a log file, `mapapp-log-<date and time>.txt`, to the same folder. The log keeps every message and the parameters of each reconstruction, registration and fit. Logs older than the newest 50 are removed at start.

6. Basic operation



The Dashboard

The MAPapp program operates from a single window with 5 panels.

Panel 1: Dashboard. The task buttons and the options that go with them. The tasks run from top to bottom; registration and masking can be left out. A green light shows a task done, yellow a task running, and red one not done or failed. At the bottom of the Dashboard one progress bar shows the progress of every task ("Registration 40 %", "Fit 40 %", "Finished"). The Fit button names what it fits: **Fit T_1** for a T_1 scan, **Fit T_2** for a T_2 scan, **Fit T_2*** for a T_2* scan, **Fit $T_{1,2}$** for an ORACLE scan, **Fit CEST** for a CEST or MT scan.

Panel 2: Raw images / Reco parameters / Fit parameters / Method. The images and the mask, and the reconstruction and fit settings. Once a scan is loaded, a **Method** tab holds the options of the mapping it allows: the water & fat methods, the QSM methods, the CEST/MT options (section 9), a diffusion scan's fit model, or the ORACLE fit's options (step 5). A gradient-echo scan that allows both water & fat and QSM gets two, **Method 1** (water & fat) and **Method 2** (QSM). Before a scan is loaded, and for a scan without such options, there is no Method tab.

Panel 3: Sequence parameters / App settings. The relevant acquisition parameters, and the app's settings (theme, sounds and updates). The count of the scan's points in the Sequence parameters tab is named after the kind of scan: **#echoes**, **#TIs**, **#TRs**, **#angles** (flip angles) or **#cycles** (phase cycles) or **#offsets** (saturation offsets). For a T_2 or T_2* scan the first echo time (**TE0**) and the echo spacing (**ESP**) can be edited; changing them after a fit clears the maps. For an ORACLE scan the echo time (**TE**) is shown, read only; ORACLE needs TE = TR/2, and the message box warns when the scan's TE differs. For a T_1 scan **TE** is shown read only too: the TE the scan was set to, else the one in the scanner's own DICOM images of the scan, else the shortest TE the sequence's timing allows (0 when none of these is

known). Beside **TR** a T_1 scan also shows its series: **angles (deg)** (the flip angles of a variable flip angle scan), **TIs (ms)** (an inversion recovery) or **TRs (ms)** (a saturation recovery); a long series shows in full in the field's tooltip.

Panel 4: the map tabs and Metadata. Only the map tabs of the loaded scan are shown; the **Metadata** tab (step 9) is always there, at the right end:

Scan	Map tabs
T_1 (inversion recovery, saturation recovery, variable flip angle)	T_1 map, M_0 & R^2
T_2 (multi-echo spin echo)	T_2 map, M_0 & R^2
T_2^* (multi-gradient echo)	T_2^* map, M_0 & R^2, Water & Fat when **water & fat** is ticked, FF & R2* when **water & fat** is ticked with a method other than B-spline, Susceptibility when **QSM** is ticked
ORACLE (phase-cycled bSSFP)	T_1 map, T_2 map, M_0 & R^2, Off-resonance
CEST or MT (saturation offsets)	Z / MTR, Asymmetry, Fitted pools, B0 & quality

Panel 5: Messages. The program messages, newest on top, and the status: OK, BUSY, WARNING or ERROR.

The mouse scroll wheel steps through the slices: over the raw images those of the images, over the maps those of the maps. Over a slider the wheel moves that slider, one step per notch.

Step 1: Loading data

Press **Load MRD** to load the MRD raw data file. On the scanner computer (Windows) the first dialog of a session opens in the newest exam of the scanner's MRD folder, unless the settings name a folder (`mrdImportPath`); elsewhere it opens in the folder of the last file loaded.

MAPapp tells the kind of scan from the MRD parameters:

- **CEST or MT**: a scan whose experiments step through saturation offsets, or one saturated image at one offset (one scan of a set, section 9);
- **T_1**: an inversion recovery (Look-Locker FLASH), a saturation recovery (an array of TRs) or a variable flip angle series;
- **ORACLE**: a 2D or 3D balanced SSFP scan repeated with different RF phase increments (at least three);
- **T_2 or T_2^***: a scan with more than one echo; a multi-gradient echo is fitted for T_2^*, a multi-echo spin echo for T_2.

A scan that is none of these is refused, with the reason in the message box; so is a cine, whose frames are no relaxation series.

EPI scans. A scan read out by echo planar imaging (EPI) is recognized from its parameters and reconstructed automatically when it is loaded; nothing needs to be set. The rest of the steps work as for any other scan. **CS reco** does not apply to an EPI scan; the message box says so. A diffusion-weighted EPI scan is not a relaxation scan: it is refused, and the message names the app that maps it. Undersampled (compressed sensing) gradient echo scans are sorted with their trajectory table, the `.txt` file beside the MRD file. Retrospective's and P2ROUD's MRD exports load as well.

For multi-receiver coil data, select one of the MRD files; the app loads the others. The relevant acquisition parameters are shown in panel 3.



The Sequence parameters of a multi-echo spin echo scan

Echo times. For a T_2 or T_2* scan **TE0** and **ESP** are filled in when the scan is loaded. When the message box asks you to, check them before fitting.

ORACLE scans. The message box says how many phase cycles the scan has, and the flip angle the scan was acquired with. Experiments at the start of the scan that only bring the signal to its steady state are left out. ORACLE needs N phase increments in equal steps round the whole circle (for 5 cycles: 0, 72, 144, 216 and 288°). When the increments cover only part of the circle, or their steps are not equal, a warning says so and gives the increments that would be needed; the scan is fitted all the same. When the exam also holds a flip-angle map scan (an AFI scan) that covers the scan's whole field of view, the message box says so and the fit uses it (see **use flip angle map** on the ORACLE Method tab, Step 5); when it holds one that does not cover it, the message box says why. The flip-angle map scan may have another matrix, field of view, orientation or position. When it was made with another RF pulse or slab thickness, a warning says what differs, and the map is used all the same.

Variable flip angle scans. As for an ORACLE scan, the message box says when the exam holds a flip-angle map scan that covers the scan, and the fit then uses the flip angles each voxel really got (see **T_1, variable flip angle** in section 7). A scan acquired with the scanner's steady state setting on used its first flip angle for every image, and a scan acquired with balanced gradients is no spoiled gradient echo series: both load with a warning, and the fit refuses them. A scan acquired with RF spoiling off loads and fits with a warning that its T_1 may be biased; the warning is kept with the exported maps.

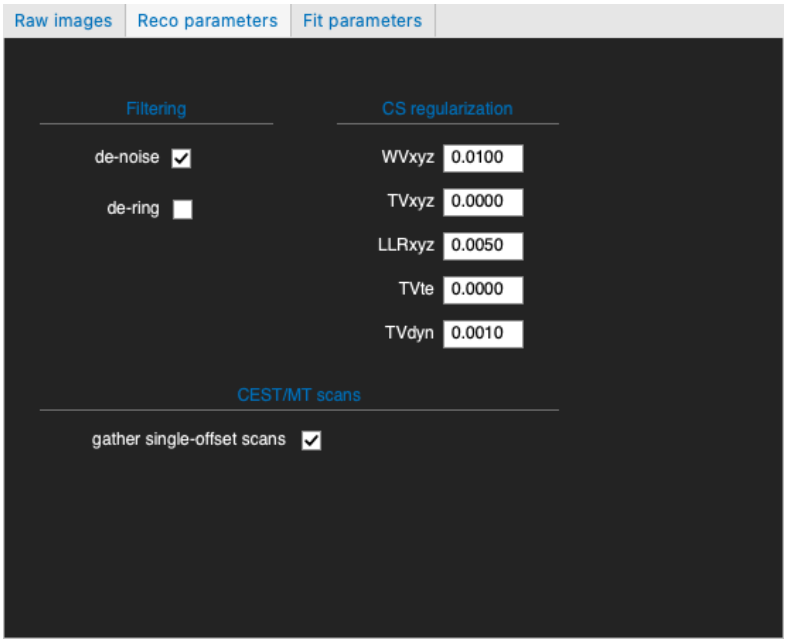
Step 2: Image reconstruction

Loading reconstructs the images already. Press **Reco images** to reconstruct them again, for instance with compressed sensing. While a reconstruction runs, the button reads **Stop**; pressing it stops the reconstruction and leaves the earlier images and the parameters as they were.

The images, the maps and every export are oriented as the scanner's own images of the scan.

The standard reconstruction is a 2D or 3D FFT with PCA de-noising, on by default. When the **CS reco** checkbox on the Dashboard is checked, the images are reconstructed by compressed sensing with the weights of the Reco parameters tab: wavelets (**WVxyz**), total variation (**TVxyz**) and locally low rank (**LLRxyz**) in space, and total variation along the points of the scan (**TVte**: the echoes, inversion times, TRs or flip angles) and along the dynamics (**TVdyn**). For an ORACLE scan, and for a CEST or MT scan, the two weights along the points and the dynamics are off: the profile over the phase cycles, or the Z-spectrum over the offsets, is what the fit reads. Compressed sensing runs with BART when it is installed, else with sigpy; the message box says which engine ran.

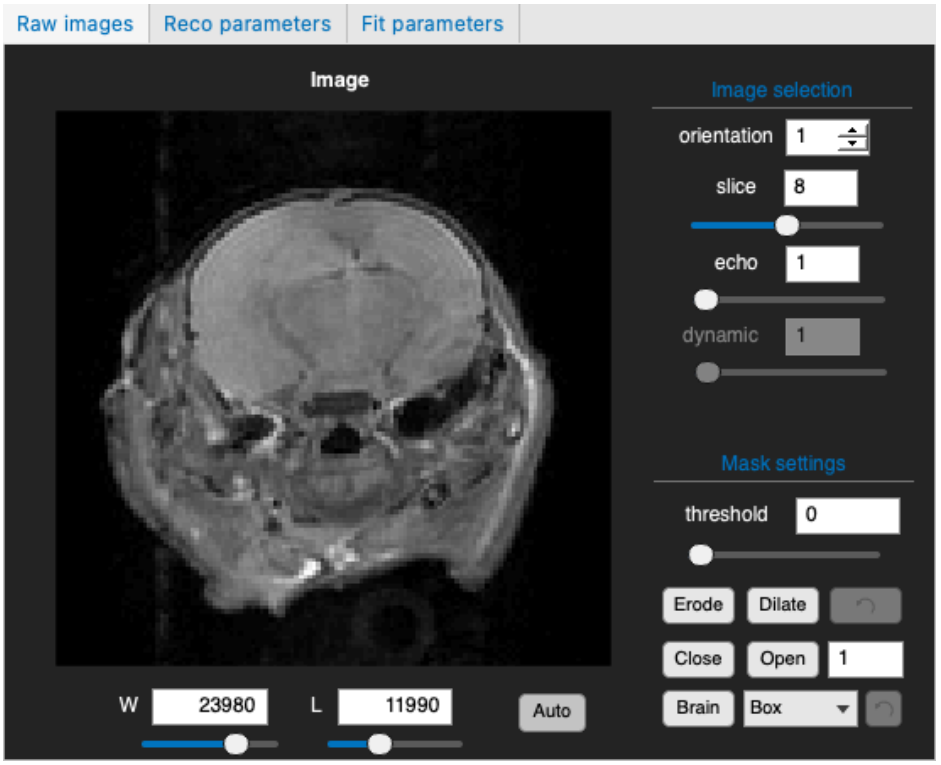
That tab also switches PCA de-noising (**de-noise**) and Gibbs ringing suppression (**de-ring**, off by default), which removes the ripple beside sharp edges.



The Reco parameters tab

The images can be inspected with the sliders of the Raw images tab: **slice**, the point of the scan (**echo** for a T_2 or T_2^* scan, **point** for the inversion times, TRs or flip angles of a T_1 scan, **cycle** for the phase cycles of an ORACLE scan, **offset** for the saturation offsets of a CEST or MT scan) and **dynamic**. The window and level are set with the **W** and **L** fields and sliders, or with **Auto**.

orientation chooses the plane the raw images show: **1** the slices as they were acquired, **2** and **3** the two planes across them, built from the slices. In planes 2 and 3 the top of the magnet is at the top of the image, and the pixels keep their true shape, so thick slices show as wide pixels. The **slice** slider then steps through the slices of that plane. The orientation is available for scans with more than one slice.



The Raw images tab

Step 3: Image registration

NOTE: this step usually can be omitted.

In case the images of the different points of the scan are not perfectly registered, press **Register** to register them as a **Translation**, **Rigid**, **Affine** or **B-Spline** transform (the list below the button). While the registration runs, the button reads **Stop**; pressing it stops the registration and leaves the images as they were. Registering clears the mask and the maps.

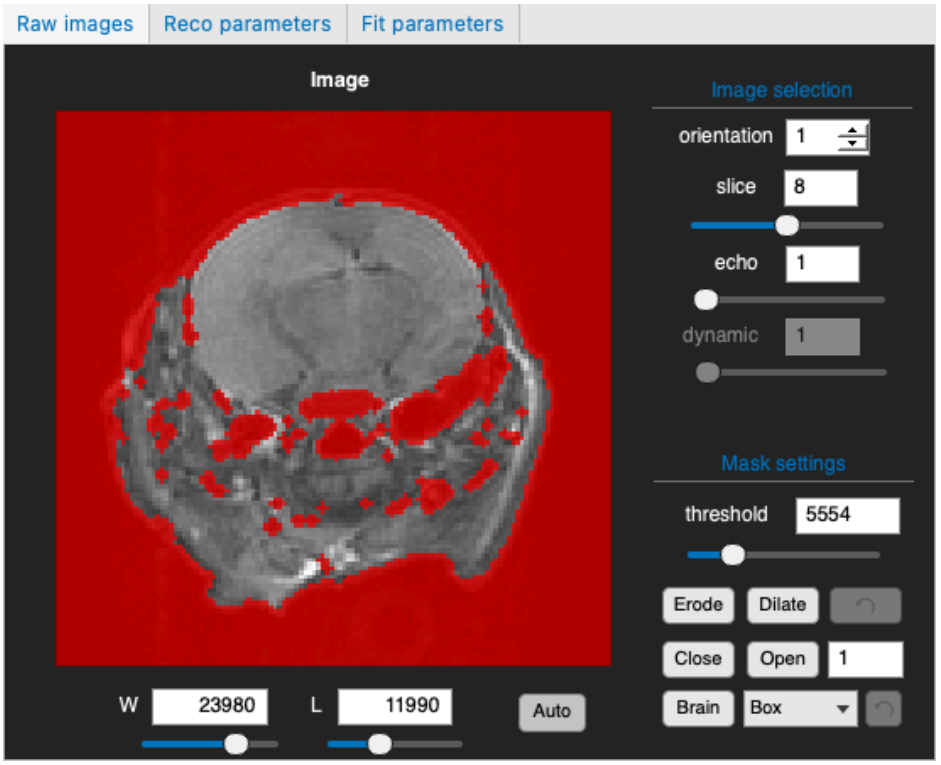
Step 4: Masking the background

To prevent unnecessary fitting of the background pixels, the images need to be masked first; without a mask every pixel is fitted, the background included.

Press **Mask** to mask them automatically. The red pixels indicate the regions that will be skipped during fitting. The threshold is found per slice for a 2D scan and once for the whole volume of a 3D scan (an ORACLE scan is masked on the image averaged over the phase cycles, a CEST or MT scan on the mean of its unsaturated reference images). The outermost layer of the object is left out, as are small specks of noise above the threshold; a dark region inside the object, such as air or a sinus, is skipped as well.

In case the automatic mask is not satisfying, the threshold can be set with the **threshold** field or slider; a threshold set by hand masks exactly the pixels above it, without the cleaning of the edge and the specks. For a 3D scan, a threshold typed or set with the slider applies to every slice; for a 2D scan it applies only to the slice on screen, and the other slices keep their automatic thresholds. Either way it applies to every dynamic. This holds for every kind of scan. The mask can be refined by **Erode**, **Dilate**,

Close and **Open** operations, of the size in pixels in the field beside them; **Undo** takes the last one back. Any change to the mask after a fit, **Mask** included, withdraws the fitted maps: fit again to see and export them.



The mask: the red pixels are not fitted

The fit region. The fit can be held to a part of the images, for every kind of scan. Only the pixels of the mask inside the region are fitted: the background inside it stays out, and outside it everything shows red.

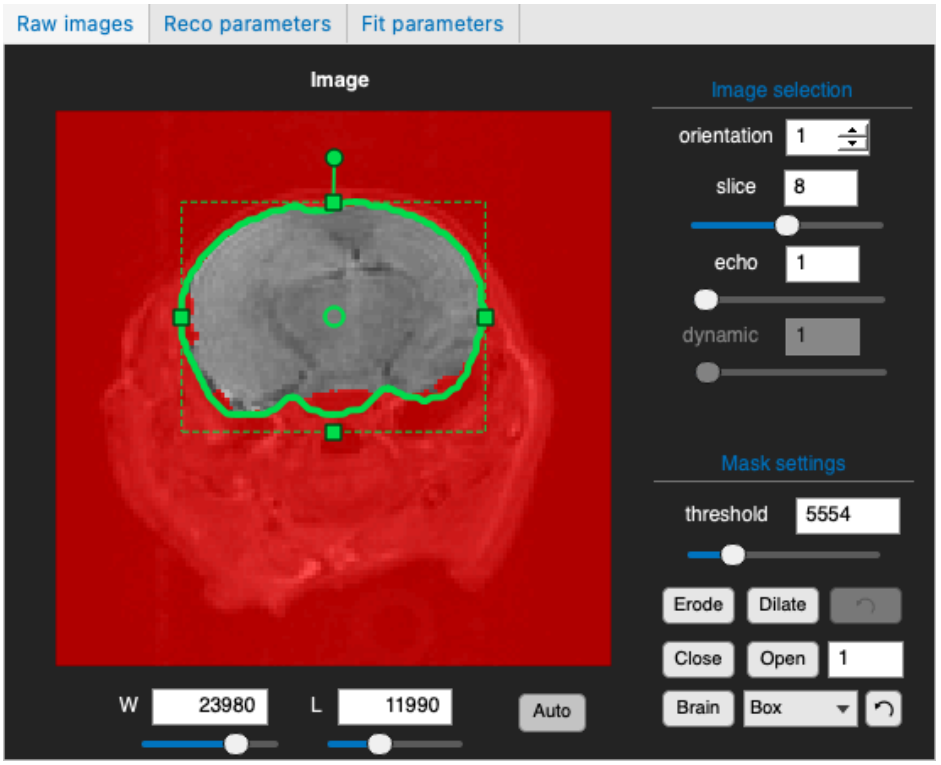
- **A box**: drag with the left mouse button on the raw image. The box runs across the plane shown as deep as the shorter side of the rectangle drawn, centered on the slice shown. Draw it in plane 2 or 3 to place it along other axes.
- **Box** / **Ellipsoid** (the list under the mask buttons): what a box fits, the box itself or the ellipsoid that fits inside it. The choice is kept for the next start.
- **Brain**: finds the brain of a mouse or rat in the images and fits only there. It looks at all of the scan's images together (every echo, inversion time, phase cycle, reference or b-value, and every dynamic), whichever one is shown, and works on a thin slab of slices too. It takes a few seconds; while it runs the button reads **Stop**, and pressing it stops the search and leaves the region as it was. The message box says the brain's volume and how well the template fits; with a poor fit it warns you to check the outline or draw a box instead. A box drawn later replaces the brain, and the brain replaces a box.

The region is drawn as a thick green line on the raw images and on the maps: solid on the slices it runs through and dashed on the others; the brain as its outline. On the raw images it has handles, in whichever plane is shown:

- a **square** on each side: drag it to make the region larger or smaller in that direction; the opposite side stays where it is;
- a **dot** above the region: drag it round to turn the region about its center, in whole degrees;
- a small **open circle** at the center: drag it to move the whole region.

The outline follows the mouse while you drag; the region changes when you let go. A brain can be stretched, turned and moved in the same way. The **Undo** button beside the list takes the last change of the region back (a second press does it again); a double-click on the raw image removes the region, and the whole mask is fitted again. A change of the region withdraws the fitted maps: fit again, and only the region is fitted.

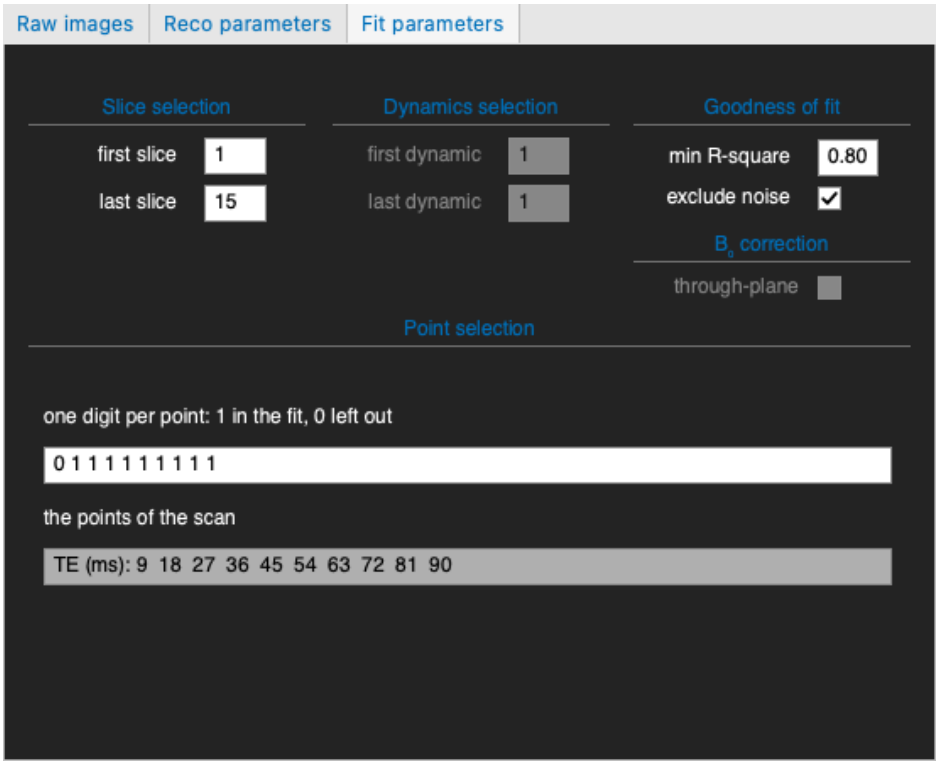
The handles of a brain sit on a thin dashed box around the whole brain, so on a slice near the brain's front or back they can lie away from its outline.



The brain found by Brain as the fit region: its outline, the squares that stretch it, the dot that turns it and the open circle that moves it; outside the region everything is red

Step 5: Fit parameters

The Fit parameters tab holds the fit settings.



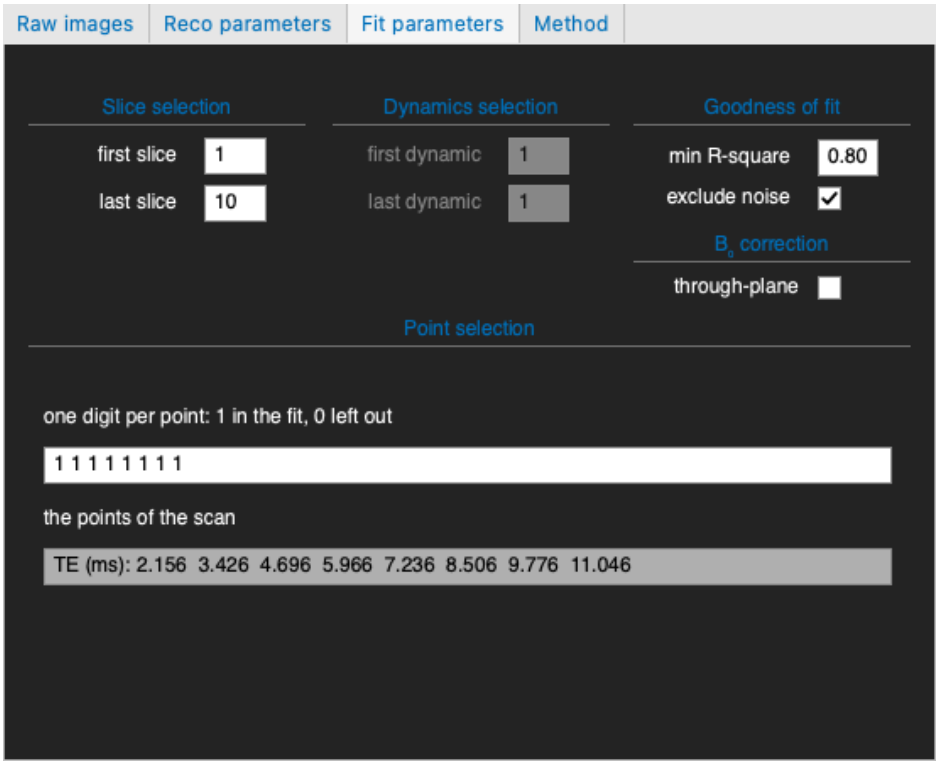
The Fit parameters tab of a multi-echo spin echo scan

Slice and dynamic selection. The first and last slice and dynamic that will be fitted (for a variable flip angle series, the cardiac frames take the place of the dynamics).

min R-square is the lowest R^2 accepted: where the fit of a pixel has a lower R^2 (0.80 by default), its values are set to zero in all maps.

Point selection. One digit per point of the scan, separated by spaces: **1** for a point in the fit, **0** for a point left out. `0 1 1 1 1 1 1 1` leaves the first of eight echoes out and fits the other seven. The line under the field lists the points in the same order: the echo times (TE), the inversion times (TI), the TRs, the flip angles, the phase increments or the saturation offsets (ppm; **S0** for a reference). The field takes only 0s and 1s and is corrected when you press Return or leave it: other characters are removed, missing digits at the end become 1, digits beyond the scan's points are removed, and when fewer points are left in the fit than it needs (2 for a T_2 fit or a CEST/MT analysis, 3 for a T_1 fit) all points are set back to 1. The message box says what was corrected. As the scan is loaded, every point is in the fit, except the first echo of a multi-echo spin echo. For an ORACLE scan the field is off: every phase cycle is fitted.


Raw images
Reco parameters
Fit parameters
Slice selection
first slice 1
last slice 4
Dynamics selection
first dynamic 1
last dynamic 1
Goodness of fit
min R-square 0.80
Point selection
one digit per point: 1 in the fit, 0 left out
1 1 1



The Fit parameters tab of a multi-gradient echo scan

Point selection for ORACLE scans. The point field on the Fit parameters tab has one digit per phase cycle: 0 leaves that cycle out of the fit, as for the echoes of a T_2 scan. At least 4 cycles stay in; with fewer, all are put back and the message box says so. The pixel's plot shows a cycle left out in red; the fit, the pixel's values and the drift correction use only the cycles chosen.

The ORACLE Method tab, for ORACLE scans, holds the options of the ORACLE fit in three groups. Every option but **α** and **AFI scan** is kept for the next start. Changing any of them after a fit withdraws the maps (the Fit lamp turns red) until the Fit button is pressed again; the B_1 map tab then shows the flip angle map that fit used.

Flip angle:

- **α (°)**, the flip angle the fit uses, from 1 to 180°. Loading a scan fills in the flip angle the scan was acquired with (15° when the scan's files do not give it); it can be changed before fitting. T_1 depends strongly on the flip angle, so check it before fitting.
- **use flip angle map**, on by default. The flip angle a voxel really gets varies over the volume. When the exam holds a flip-angle map scan (an AFI scan) that covers the scan's whole field of view, the fit uses it: each voxel gets **α** times what the map measured there. The message box says how far the flip angle differed from **α** over the mask. When the map covers too little of the mask, or its values are not plausible, the message box says so and the fit uses **α** everywhere. Untick it to fit with **α** everywhere.
- **AFI scan**: which flip-angle map scan is used, among those of the exam that cover the scan, the nearest in scan number first (**none** when the exam has none).
- **map smoothing (voxels)**, 1.5 by default, from 0 to 5: how much the flip-angle map is smoothed before it is used, in voxels of the flip-angle map scan.

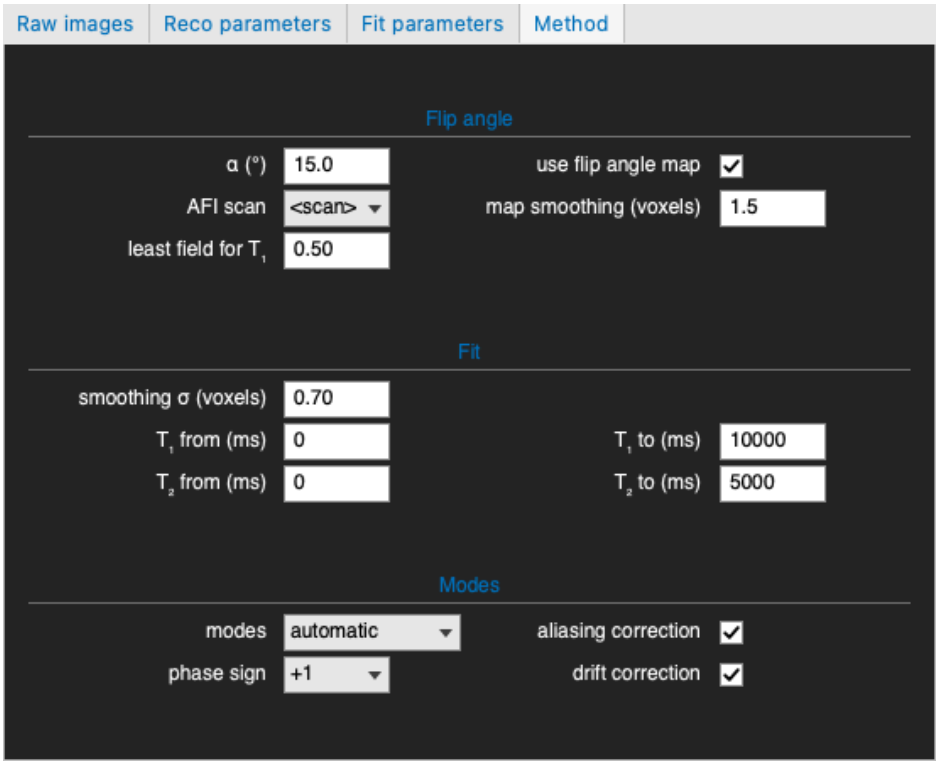
- **least field for T_1**, 0.5 by default, from 0 to 1: where a voxel got less than this fraction of **α** (the edges of a slab), it gets no T_1 or M_0 (T_2 and the off-resonance do not depend on the flip angle and are kept).

Fit:

- **smoothing σ (voxels)**, 0.7 by default, from 0 to 5. The T_1 and T_2 maps of a noisy scan are steadier when the fit smooths over the neighboring voxels inside the mask, over a width of σ voxels; 0 fits every voxel on its own. The off-resonance follows each voxel's own data.
- **T_1 from / to (ms)** and **T_2 from / to (ms)**: the T_1 (0 to 10000 ms by default) and T_2 (0 to 5000 ms) taken as a fit; a voxel outside them keeps 0 in every map. **min R-square** on the Fit parameters tab applies as for every scan.

Modes:

- **modes**: how the fit reads the profile. **automatic** (the default) takes the averages over the phase cycles when the phase increments are equal steps round the whole circle, and a least-squares fit otherwise; **means** always the averages; **least squares** always the fit.
- **aliasing correction**, on by default: corrects the averages for the effect of having only a few phase cycles (it applies to equal steps round the circle).
- **phase sign**: the scanner's phase convention, **+1** (the default, right for the scanner the examples come from) or **-1**. With the wrong one the T_1 map comes out mostly empty.
- **drift correction**, on by default. The scanner's frequency can drift a little during the minutes the phase cycles take, which moves the later cycles' signal. The fit then finds, from the scan itself, how far each phase cycle was moved, and fits with that. The message box gives the shifts found (in degrees and in Hz) and how much better the fit became; when the correction does not improve the fit clearly, the message box says so and the fit is the one without it. Untick it to fit without it.



The ORACLE Method tab

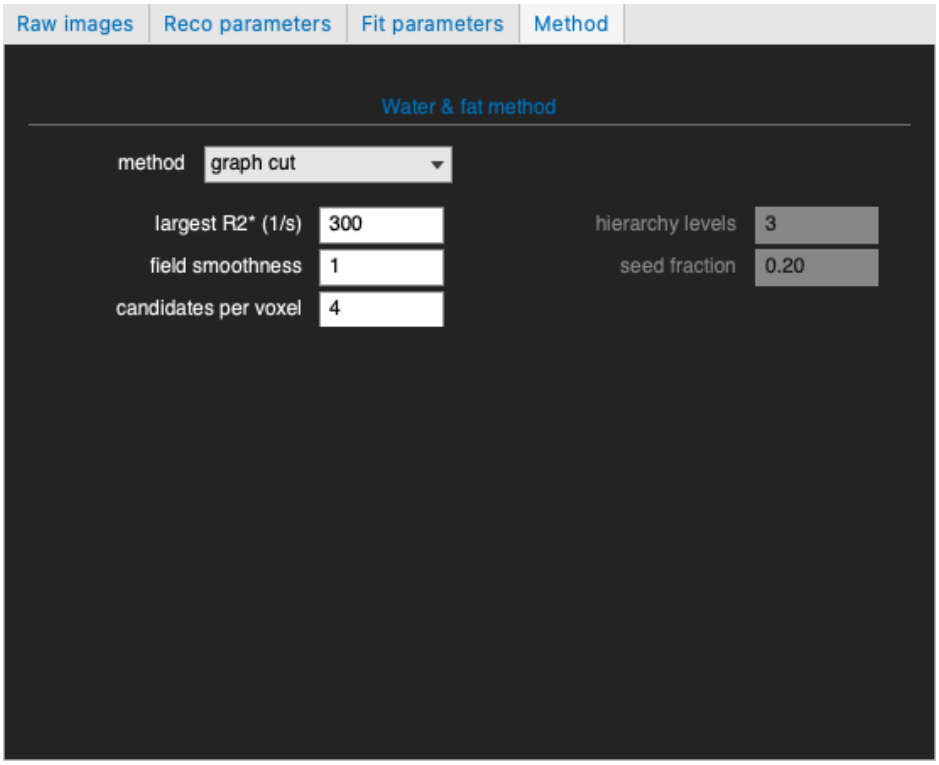
Water & fat, on the Dashboard: for a multi-gradient echo scan, tick **water & fat** before fitting to also separate water from fat. It needs at least 3 echoes, echo times at which fat and water can be told apart, and a scan without fat suppression. For a scan that does not allow it the box is off, and the message box says why when the scan is loaded. Echo times that tell fat from water only barely (an echo spacing close to a whole number of in-phase periods, section 8) leave the box on, with a warning in the message box at the load and at the fit: the maps can then be noisy and show patches where water and fat are swapped. The method is chosen in its Method tab (below).

QSM, on the Dashboard: for a multi-gradient echo scan the Dashboard shows a **QSM** tick in the place of **model-based**. Tick it before fitting to also make a map of the magnetic susceptibility χ (step 7). It is offered for 3D multi-gradient echo scans with at least 3 echoes whose files give the imaging frequency and the direction of the main field; for other scans the tick is off. The susceptibility map takes some tens of seconds on top of the T_2^* fit. An undersampled (compressed sensing) scan must be reconstructed with **CS reco** first. The QSM methods are chosen in their Method tab (below). For QSM, a scan of one average is the safer protocol; for a scan of several averages the message box says so.



The Dashboard of a 3D multi-gradient echo scan, with QSM ticked

The water & fat and QSM Method tabs hold the methods of the water & fat separation and of QSM, with their parameters, one tab each; a scan that allows only water & fat shows only its tab. Only the fields of the chosen methods are on; every choice and value is kept for the next start.



The water & fat Method tab

Water & fat method, in the upper half:

- **graph cut** (preferred, the standard choice): chooses the field map for the whole slice at once, weighing each pixel's fit against the smoothness of the field; it leaves the fewest patches where water and fat are swapped.
- **hierarchical IDEAL**: finds the field map from coarse blocks down to single pixels; the fastest, with a little more speckle.
- **region growing**: grows the field map from the most reliable pixel outwards to its neighbors.
- **B-spline**: a smooth field map over the slice, without R_2* and without the correction of the bipolar readout.

Graph cut, hierarchical IDEAL and region growing fit R_2* together with water and fat, and correct the phase difference between the odd and even echoes of a bipolar readout. Their parameters: **largest R2* (1/s)**, the highest decay rate searched (300 by default; 0 leaves R_2* out); **field smoothness**, how strongly neighboring pixels are held to similar fields (1 by default); **candidates per voxel**, the number of possible fields each pixel offers (graph cut, 4); **hierarchy levels**, the number of block sizes (hierarchical IDEAL, 3); **seed fraction**, the share of the brightest pixels the starting pixel is chosen from (region growing, 0.2).

QSM method, in the lower half: the **background** field removal (**LBV**, the standard choice, which keeps the tissue up to the mask's edge; **V-SHARP**; **PDF**) and the **inversion** that turns the field into χ (**TV (ADMM)**, the standard choice, with the least streaking; **TKD**, a quick preview; **closed-form L2**; **LSQR**). Their parameters: **V-SHARP radius (mm)**, **PDF tolerance**, **TV weight** and **TV iterations**, **TKD threshold**, **L2 weight**, and **iLSQR tolerance** and **iLSQR iterations**.

model-based, on the Dashboard (T_1 and T_2 scans): tick it to fit a model-based reconstruction with BART, the maps then being fitted to the k-space directly. It is offered when BART is available, for multi-echo spin echo scans and for 2D inversion recovery scans (not for saturation recovery, flip angle or 3D T_1 scans); an inversion recovery whose images were registered is fitted pixel-wise. Its R^2 map tells how well the fitted curve follows the images; min R-square is not applied.

Step 6: Fitting the maps

Press the Fit button (**Fit T_1**, **Fit T_2**, **Fit T_2***, **Fit $T_{1,2}$** or **Fit CEST**) to start the fit. While the fit runs, the button reads **Stop**; pressing it stops the fit and leaves the maps as they were before it. The maps fill in slice by slice while the fit runs. After the fit the maps show the middle slice.

A press that would give the same maps again, with nothing changed since the last fit, is skipped with a message in the message box; the same holds for **Reco images** and **Register**. When only an extra is new, for example **QSM** ticked after a fit, only that extra is calculated and the maps are kept. With **water & fat** ticked and a method that fits R_2* (all but the B-spline method), the T_2* map comes from the water & fat fit (section 7) and the single-exponential fit is left out.

What each fit calculates is described in section 7. For every kind of scan, pixels outside the mask, pixels whose R^2 is below min R-square and pixels with a value outside the possible range (T_1 above 10 s, T_2 above 5 s) are zero in all maps.

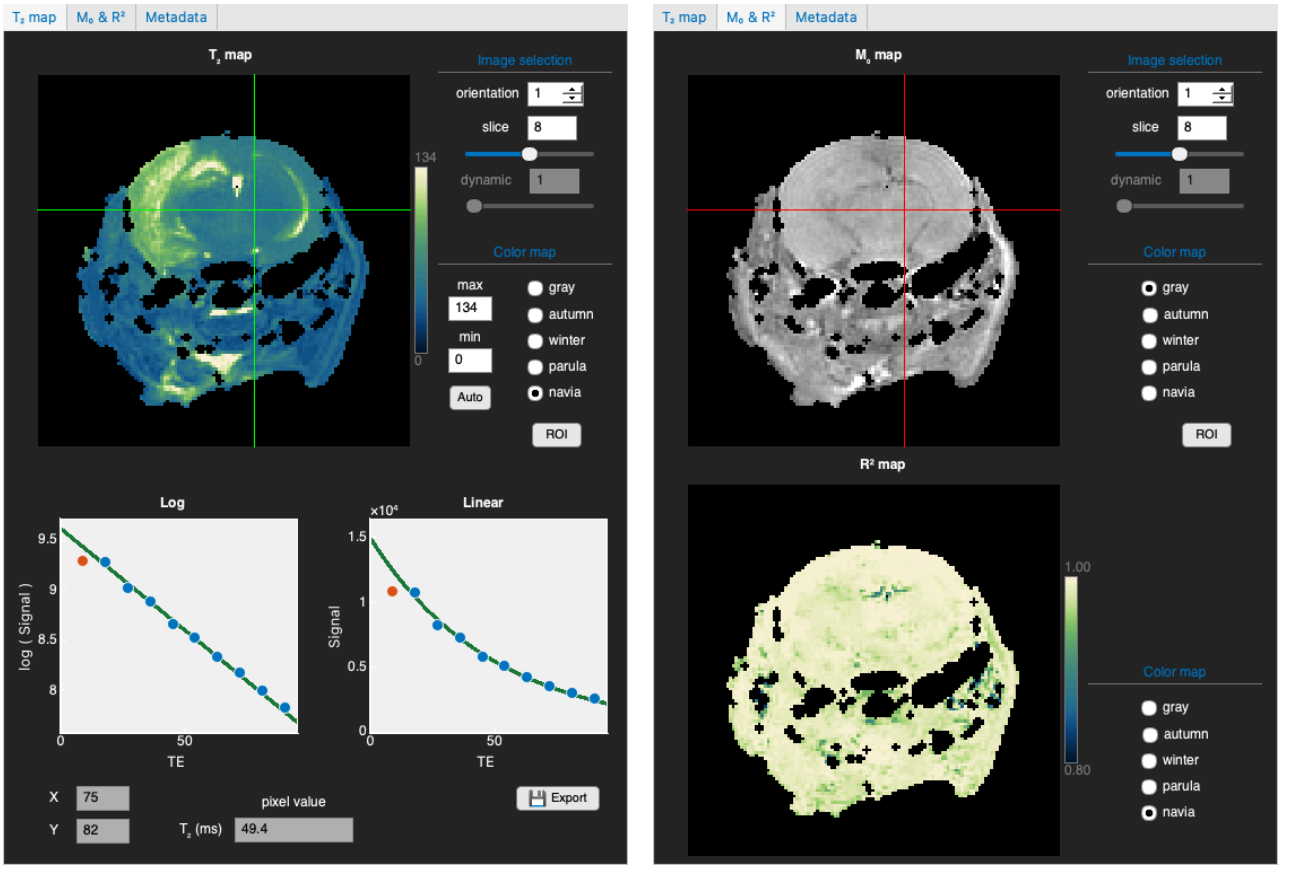
Step 7: The map tabs

Each map tab shows a map on a scale set with **max** and **min**, or with **Auto**, and a color map chosen beside it (gray, autumn, winter, parula and navia; for the T_1 map lipari in the place of navia). The color bars stand beside the maps with their end values above and below them; a value too long for the room beside the bar is written shorter (fewer digits, or as -6e-3 for -0.006), the **max** and **min** fields giving it in full. The slice and dynamic sliders of the map tabs move together: every tab shows the same slice and dynamic.

orientation chooses the plane the maps show, as for the raw images (step 2): **1** the slices as acquired, **2** and **3** the planes across them. Every map tab shows the same plane, and its slice slider steps through that plane's slices. A fit region is drawn on the maps as on the raw images.

Auto sets the scale from the values of the slice shown: the T_2 map from 0 to twice its mean T_2; the T_1 map of a T_1 scan from 0 to twice the median T_1; the T_1 map of an ORACLE scan from 0 to twice its mean T_1; the off-resonance map over its whole range (below).

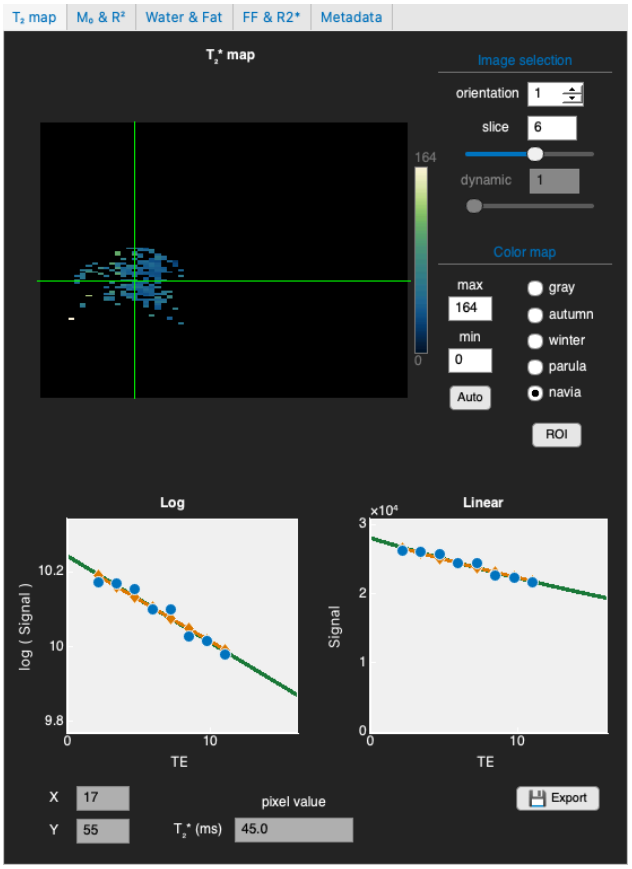
T_2 map **M_0 & R^2**



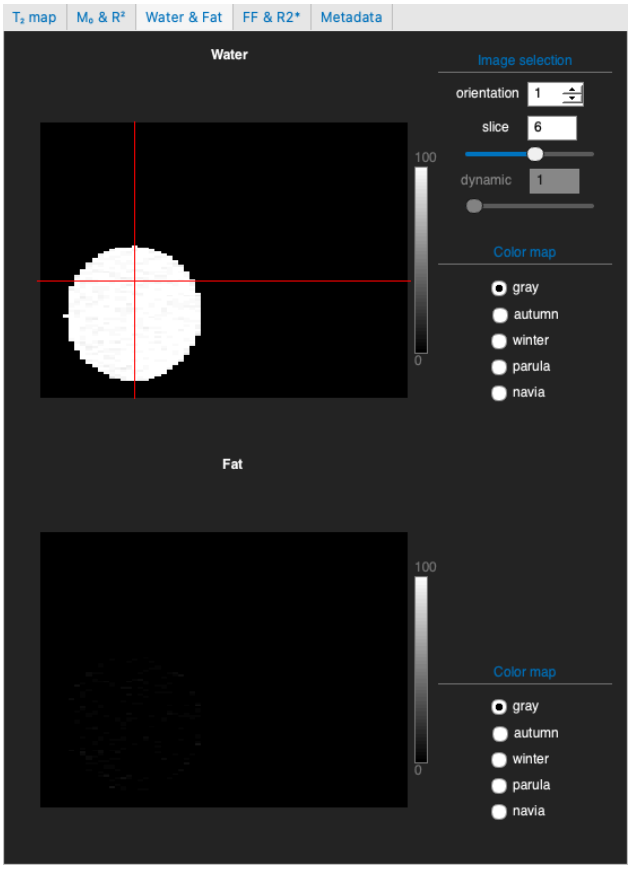
M_0 & R^2: the M_0 map (the signal the fit gives at time zero, or the profile's scale for ORACLE) and the R^2 goodness-of-fit map, its color bar from the R^2 cut-off to 1.

T_2* and water & fat. Multi-gradient echo scans are fitted in the same way as spin echo scans; the maps, the fields and the plots then say T_2*. With **water & fat** ticked, the Water & Fat tab shows the water and fat maps in percent of the total signal, each with its own color map and a color bar from 0 to 100 %.

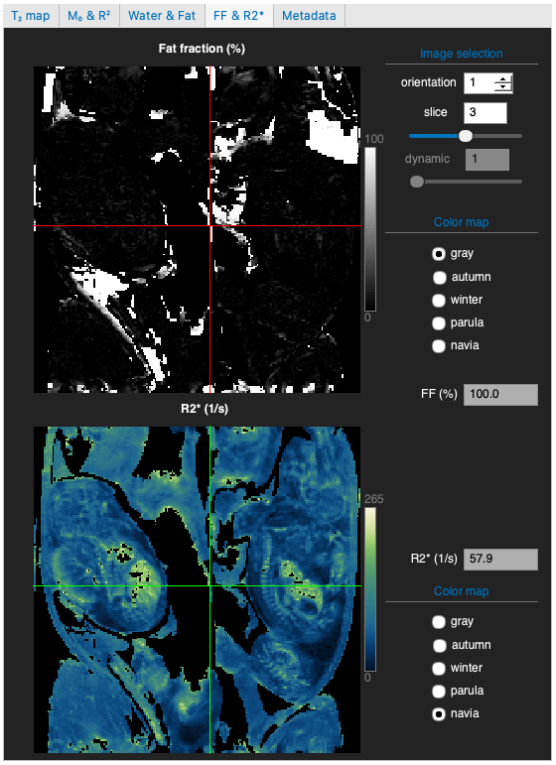
T_2* map



Water & Fat

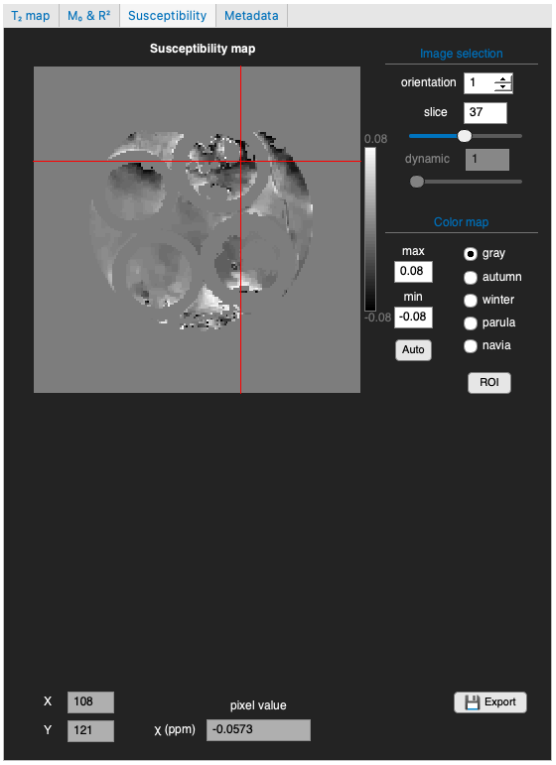


FF & R2* (with **water & fat** ticked and graph cut, hierarchical IDEAL or region growing chosen in its Method tab): the fat fraction map in percent, from 0 to 100 % (gray by default), above the R_2* map in 1/s (navia by default), whose color bar runs from 0 to the largest R_2* of the map. Each map has its own color map, kept for the next start. The tab has no **max**, **min** or **Auto**; its slice and dynamic sliders move with those of the other map tabs. Extreme R_2* pixels are set to 0, and the message box says how many. A click on either map places the cross-hair (or adds to the ROI) on both; the fields **FF (%)** and **R2* (1/s)** then show the fat fraction and R_2* of the pixel, or the average over the ROI, inside the mask.



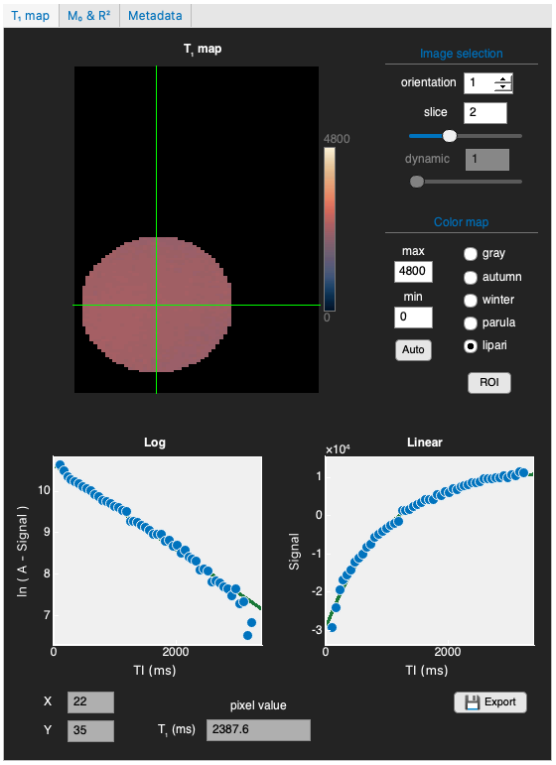
The fat fraction and R_2* maps of a multi-gradient echo scan of a mouse abdomen

Susceptibility (with **QSM** ticked): the map of the magnetic susceptibility χ in ppm, relative to the mean over the mask (the mean is zero), gray by default. Paramagnetic regions (iron, blood) are positive, diamagnetic regions (calcium, myelin) negative. Its **Auto** scale is symmetric about zero, up to the largest value of the map. Voxels where the scan's phase cannot be used are zero, and so are extreme pixels; the message box says how many. The pixel value is χ in ppm; there is no signal plot for this map.



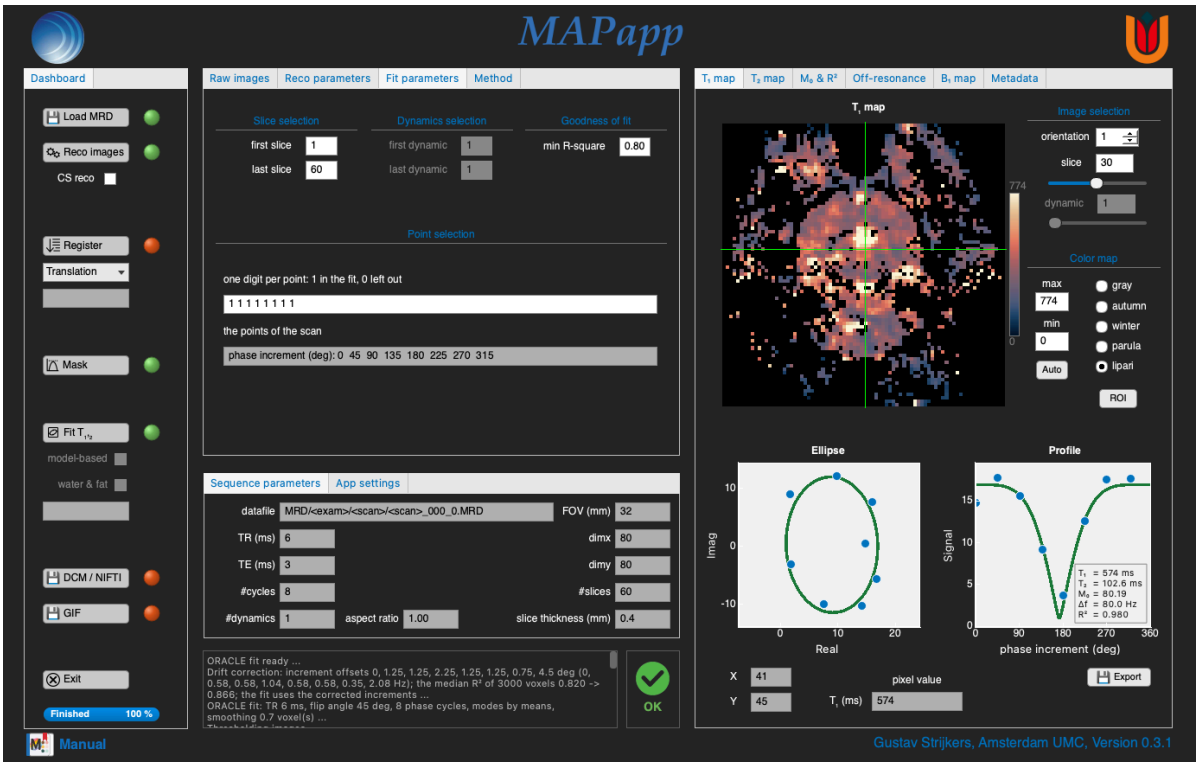
The susceptibility map of a 3D multi-gradient echo scan of a tube phantom

T_1 map of a T_1 scan: the plot below the map shows the signal along the inversion times, TRs or flip angles. The signal of an inversion recovery is shown with the sign the fit restored.



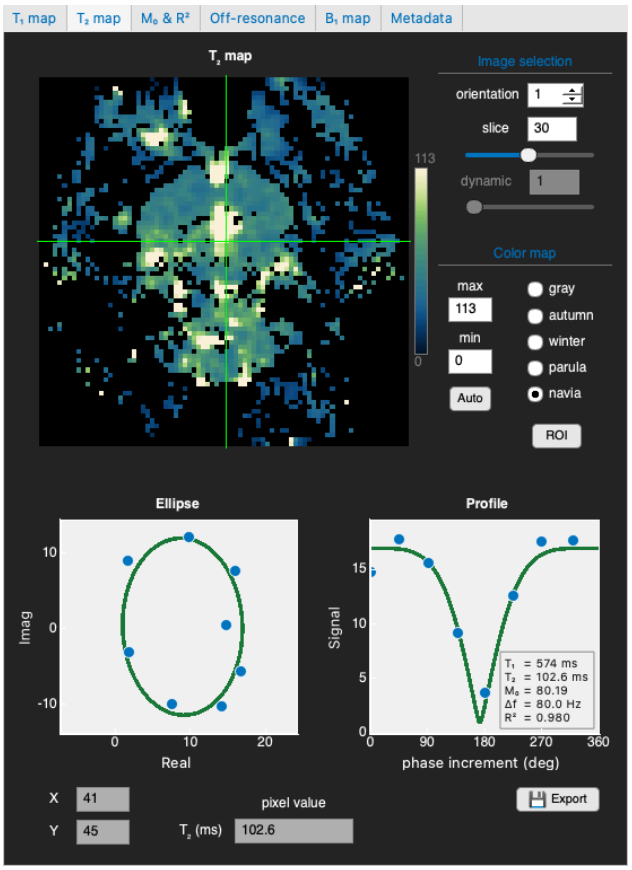
The T_1 map of an inversion recovery scan of a phantom, with the signal of a pixel along TI

ORACLE scans have four map tabs: **T_1 map**, **T_2 map**, **M_0 & R^2** and **Off-resonance**, and the **B_1 map** tab (below). The off-resonance map is in Hz, from −1/(2 TR) to +1/(2 TR): an off-resonance beyond that range shows up as one within it. Its Auto scale is that whole range.

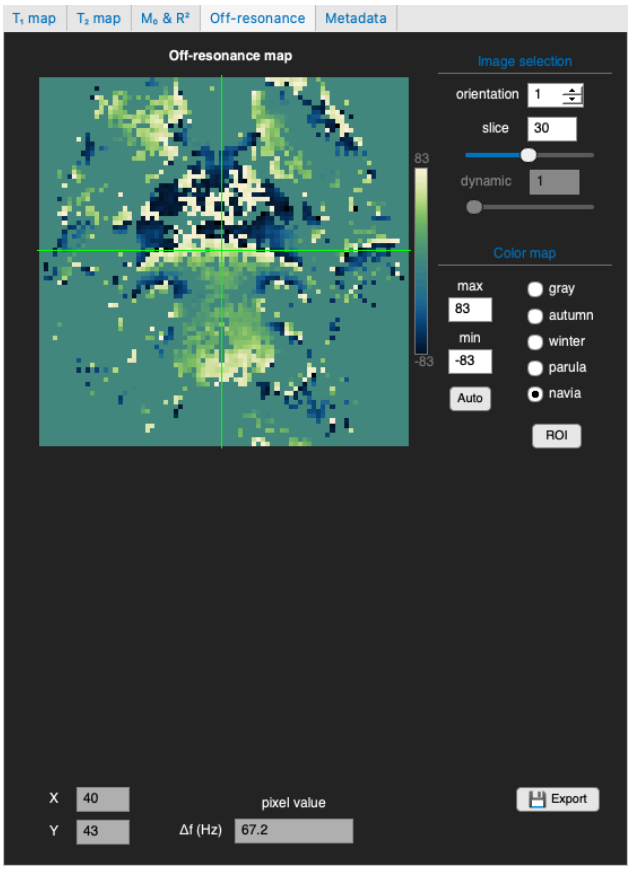


The window after fitting an ORACLE scan of a rat brain

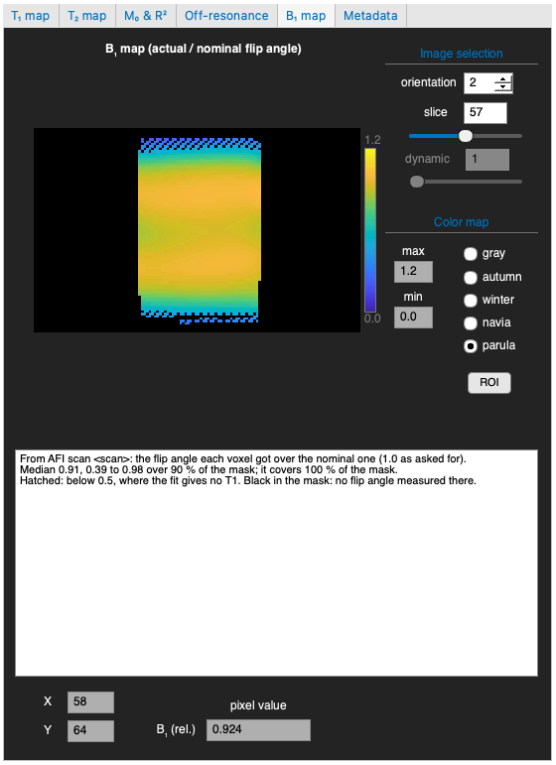
T_2 map



Off-resonance



B_1 map (ORACLE scans and variable flip angle scans): the flip angle map the fit used, from the exam's flip-angle map scan (an AFI scan), shown on the scan's own slices and inside its mask, as the other maps are: the flip angle each voxel got divided by the nominal one, so 1.0 means the flip angle was as asked for. Its scale is fixed from 0 to 1.2, with its color bar; the color map can be chosen as for the other maps. Hatched voxels (every third diagonal black) got less than half the nominal flip angle, the edges of the slab where the fit gives no T_1; black voxels inside the object have no flip angle measured. The text below the map names the flip-angle map scan, gives the median flip angle ratio and its range over the mask, and repeats any warning about the flip-angle map scan (another RF pulse or slab thickness). The slice, dynamic and **orientation** controls, the cross-hair and the ROI work as on the other map tabs; the pixel value is the flip angle ratio of the pixel, or its average over the ROI. While the fit used no flip angle map, the map stays empty and the text says so. Inversion and saturation recovery scans have no B_1 map tab.



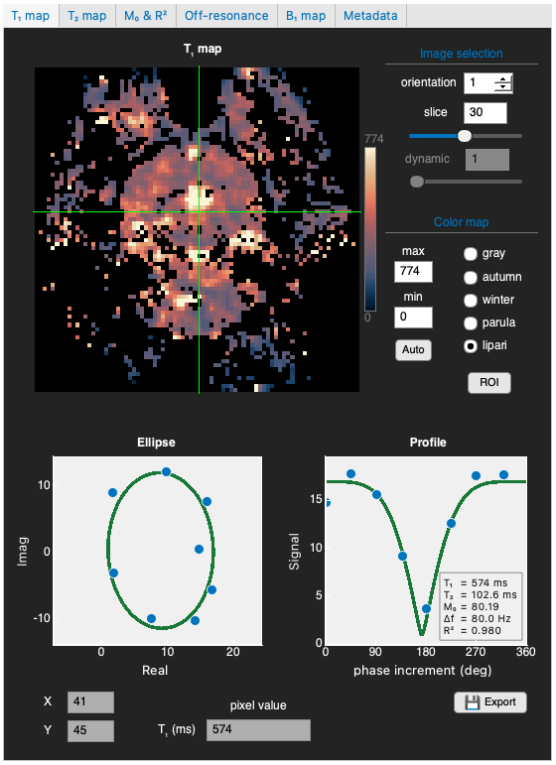
The B_1 map of an ORACLE scan, in a plane across the slab: the slab's edges hatched

Step 8: Values of pixels and ROIs

When you click on a map a cross-hair is shown, with the values of that pixel (**pixel value**) and a plot of its signal: symbols for the measured points, a line for the fitted curve. A click outside the mask shows nothing.

- **T_2 and T_2* scans**: log(signal) and signal as a function of TE. A red symbol shows an echo left out of the fit: by the point selection (step 5), because its signal was at the noise level (**exclude noise**), or because the B_0 correction left too little of it.
- **T_1 scans**: the signal along TI, TR or flip angle; a red symbol shows a point left out by the point selection.
- **ORACLE scans**: the T_1 map and T_2 map tabs show the same two plots. **Profile** (right): the size of the signal over the phase increment, 0 to 360°, with the profile that the fitted T_1, T_2, M_0 and off-

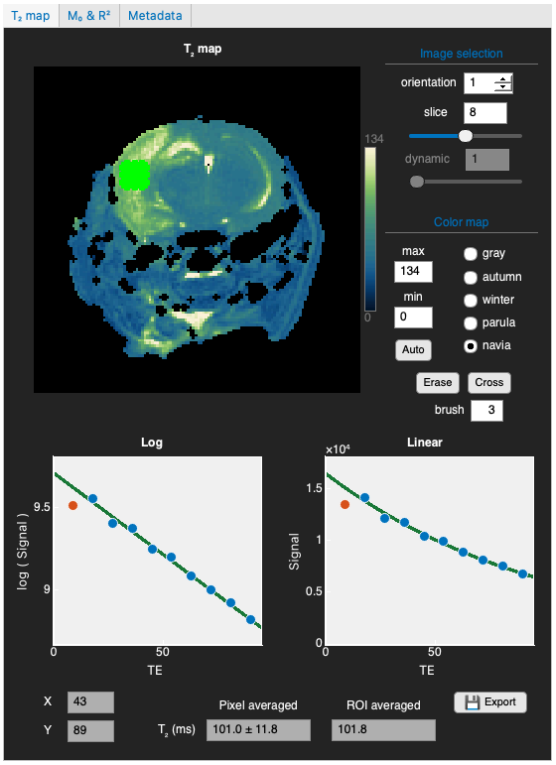
resonance predict as the line, and a box with these values and R^2 (Δf is the off-resonance), at the right edge of the plot, where it covers the fewest points. **Ellipse** (left): the measured signal as points in the complex plane (real and imaginary part, both axes in the same units), with the predicted profile as a closed line. The T_1, T_2 and off-resonance of the pixel are shown in the T_1 map, T_2 map and Off-resonance tabs.



An ORACLE pixel: its profile over the phase increments, with the fitted profile

A click works in every orientation: in plane 2 or 3 it shows the values of the voxel under the pointer, and **X** and **Y** give its place in that plane. The cross-hair and the ROI belong to the voxels, so they show in each plane on the slices that hold them.

You can toggle between the cross-hair and region-of-interest (ROI) values with the **ROI** / **Cross** button. A single ROI can be 'painted' on the map, in one or more slices, by mouse clicks on the image, with the brush size in pixels set below the button.



An ROI painted in the stroke lesion, with its signal and the fitted curve

For the ROI, two values are calculated:

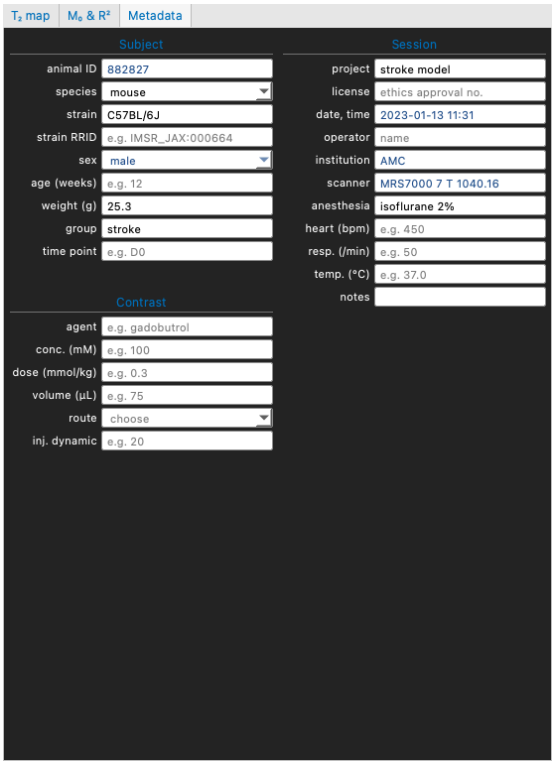
- **Pixel averaged**: the average of the values of the pixels; for T_1 and T_2 scans this is 1 / mean(1/T) over the pixels, with its spread; for ORACLE scans and for χ the mean of the map's values.
- **ROI averaged**: the model is fitted to the average signal of the ROI, producing a single value for the ROI.

The ROI can be removed with **Erase**. **Export** writes the ROI-averaged signal to a CSV file, `roiData-<tag>.csv`: the echo times, inversion times, TRs, flip angles or phase increments, beside the signal. When the MRD file lies in the scanner's folders, `…/MRD/<exam>/<scan>`, the file goes to `…/Data/<exam>/MapAnalysis/<tag><kind>/<n>`, a new numbered folder each time, without a question; otherwise the app asks for a folder. A CEST or MT scan writes its Z-spectrum, plot, figure and workbook there instead (section 9).

Step 9: Metadata

The **Metadata** tab, the rightmost of the map tabs (always shown, whichever map tabs the scan has), holds the record of the scan, which the DCM / NIFTI and GIF exports write with the maps. Its fields stand in three groups:

- **Subject:** animal ID, species, strain and its RRID, sex, age, weight, group and time point;
- **Session:** project, ethics license, date and time, operator, institution, scanner, anesthesia, heart and respiration rate, and temperature, with a field for notes;
- **Contrast:** agent, concentration, dose, volume, route and the dynamic it was injected at.



The Metadata tab: the values a load filled in (blue) and values typed in (black)

At a load the app fills in what the scan's files say: the animal ID and the time point (from the patient name typed at the console: `<date>_<animal>_D0`, for example `YYYYMMDD_123456_D0`, is animal 123456, day 0), the date and time, the institution and the scanner, and the project. What the console leaves at its standard values (sex O, weight 0, the login as the operator) is left empty.

What was filled in for an earlier scan of the same exam is filled in again: the animal's details are typed only once per exam. The date, the heart and respiration rate, the temperature, the notes and the injection's dynamic belong to one scan and are never taken from another.

Your institution, operator, anesthesia, species, strain and strain RRID are kept for later exams: they show, in blue, from the start, and a scan's own value takes their place where it has one.

A value the app filled in is shown in blue; every field can be corrected, and then counts as yours. Species, sex and route are lists to choose from, or to type in; numbers are kept with their units (a number field holding something else is edged in red). The species and sex are written with their codes (NCBI Taxonomy, PATO), so the record can be read by other software as well as by people.

Step 10: Exporting the maps

There are two ways to export the maps, and one for the ROI data.

1. **DCM / NIFTI**: exports the maps in DICOM and NIfTI format for further processing in third-party software, and the relaxation maps also as BIDS quantitative maps, in seconds (the DICOM and NIfTI maps are in ms). The DICOM files carry the species, strain, sex, age and weight of the animal where the Metadata tab has them. The exported maps lie in the scanner's geometry, at the app's own matrix, so that they overlay the scanner's images of the scan in other software.
2. **GIF**: exports the maps as animated GIF files, one frame per slice, or one GIF per slice with a frame per dynamic, in the color maps and scales shown, with pictures of the app's window: one

with the M_0 & R^2 tab and one with the T_2, T_2^* or T_1 map tab (the T_1 map for an ORACLE scan), or, for a CEST or MT scan, one of each CEST tab that shows a map, or, for a diffusion scan, one of the ADC map tab and of each IVIM tab.

3. **ROI data**: the **Export** button of the map tab (step 8).

The maps each kind of scan exports:

Scan	DICOM and NIfTI	BIDS	Folder name
T_1	T_1, M_0, R^2; B_1 for a variable flip angle scan whose fit used a flip angle map	`_T1map`; `_TB1map` (% of the nominal flip angle) with the B_1 map	`<tag>T1`
T_2	T_2, M_0, R^2	`_T2map`	`<tag>T2`
T_2^*	T_2^*, M_0, R^2; with water & fat: water, fat, and with graph cut, hierarchical IDEAL or region growing also the fat fraction (%) and R_2^* (1/s); with QSM: susceptibility χ (ppm)	`_T2starmap`; `_FFmap` (%) and `_R2starmap` with the fat fraction and R_2^*; `_Chimap` with QSM	`<tag>T2`
ORACLE	T_1, T_2, M_0, R^2, off-resonance (Hz); B_1 where the fit used a flip angle map	`_T1map`, `_T2map`; `_TB1map` (% of the nominal flip angle) with the B_1 map	`<tag>T1T2`
CEST or MT	every map of the four CEST tabs, each its own series and file, named by the map (for example `MTRasym_APT`, `MTR_p3.50ppm`), with `CEST_parameters.json` (section 9)	none	`<tag>CEST`
Diffusion	ADC, M_0, R^2, the IVIM maps after an IVIM fit, and the diffusion-weighted images (`DWI`, with `.bval` and `.bvec`), with `recoparameters_<tag>.txt` (section 10)	`_ADCmap` (mm^2/s), and the images as a dwi run	`<tag>ADC`

The GIF export writes the maps the window shows as animations, the fat fraction and R_2^* included. BIDS has no name of its own for a fat fraction map; the app names it `FFmap`, as its DICOM and NIfTI series, in percent. The B_1 map is exported to DICOM, NIfTI and BIDS only, not as a GIF.

When the MRD file lies in the scanner's folders, …`/MRD/<exam>/<scan>`, the exports go to …`/Data/<exam>/DICOM`, `Nifti`, `BIDS` and `GIF`, each into a new numbered folder; otherwise the app asks for a folder. The folder dialog opens in the folder chosen last. The folder and file names the app makes from the scan's name hold no spaces: a space, or any character other than a letter, a digit, `-`, `_` or `.`, becomes `_`.

The DCM / NIFTI and GIF exports also write, in …`/Data/<exam>/Metadata/<tag><kind>/<n>` (or under the folder asked for), a new numbered folder each time:

- `metadata.json`: the Metadata tab's record;
- the report of the export, as `.json` and `.txt`: the app and its libraries, the scan and a checksum of each of its files (for a CEST or MT scan also those of the scans read with it), its acquisition (with

the phase increments of an ORACLE scan, or the saturation offsets and pulse of a CEST or MT scan, in the place of echo times), every step of the session with its parameters, the reconstruction, registration and fit settings, the statistics of each map per slice and dynamic within the mask, the points used, for the GIF export the scales and color maps, and every file the export wrote with its checksum;

- `ro-crate-metadata.json`: the same as an RO-Crate (www.researchobject.org/ro-crate), a standard description of the folder for data repositories.

The paths in them are relative to the exam folder (or the folder asked for), so they hold no names of your own folders. The ROI data export writes no record.

Step 11: Exit

Press **Exit** to shut down the program. Closing the window does the same, also while a step runs: the reconstruction, registration or fit is stopped, the window disappears, and the program closes as soon as the step has ended.

7. What is calculated

T_2 and T_2*. Each pixel is fitted to

> *Signal = M_0 exp(–TE / T_2)*

For multi-gradient echo data the same equation gives T_2*. With the B_0 correction the equation also holds the signal lost through the slice. R^2 compares the fitted curve with the measured signal. With water & fat ticked and a method that fits R_2* (all but the B-spline method), the T_2* map is 1000 / R_2* of the water & fat fit instead, with its M_0 and R^2: one decay shared by water and fat, so that the beating of fat against water is not taken for decay.

Water & fat. The complex echoes of a multi-gradient echo scan are separated into a water and a fat signal, fat with its spectrum of several peaks, together with the field map and (except with the B-spline method) the decay rate R_2*, after the phase difference between the odd and even echoes of a bipolar readout is corrected. The maps give water and fat each in percent of the total; with graph cut, hierarchical IDEAL and region growing also the fat fraction, fat / (water + fat) in percent, and R_2* in 1/s.

Properties. Every pixel holds water and fat that share one field and one R_2*. Because R_2* is part of the model, the fat fraction does not depend on the echo times, and the several fat peaks make it accurate at low and high fractions. The fat fraction comes from the signal magnitudes, which avoids phase errors; the field map comes first, from the complex echoes.

Limitations.

- **Water–fat swaps.** Water and fat differ in frequency by about 3.4 ppm, and a field offset of the same size looks the same in a single pixel. The methods choose the field by its smoothness across the slice; where that fails (large field steps at air or bone, tissue islands apart from the rest), water and fat can swap in a patch. Graph cut swaps least; hierarchical IDEAL is fastest and shows more single-pixel swaps; region growing depends on where it starts; B-spline assumes a smooth field and swaps most at field steps.
- **One R_2* for water and fat,** and a fixed fat spectrum (that of body fat). Tissue whose fat differs, or water and fat that decay differently, give small errors in the fat fraction.
- **T_1 weighting.** Fat recovers faster than water; a large flip angle or a short TR raises the fat fraction (section 8).
- **Noise.** Where the signal is low, the fat fraction is noisy, and very low fractions read a little high.
- **Echo times.** Echoes spaced at a whole number of in-phase periods, or too few of them, cannot separate water from fat (MAPapp then does not offer it); echoes spaced close to it separate them poorly (MAPapp warns), with noisy maps and swapped patches, and R_2* from them is no better than the T_2* map.
- **Readout.** Fat lies shifted along the readout by the chemical shift (about one pixel with the bandwidths of section 8); a bipolar readout's odd and even echoes are corrected, but a strong difference between them can remain at the edges of the object.
- **2D scans** are separated slice by slice; a swap can differ between neighboring slices.
- The B-spline method has no R_2* and no correction of a bipolar readout: its water and fat are weighted by the decay over the echoes.

Susceptibility (QSM). The phase of the echoes gives the magnetic field in every voxel. The part of the field that comes from outside the object is removed, and the remaining local field is converted into the susceptibility that causes it. χ is given in ppm, relative to its mean over the mask.

Properties. Each voxel's χ is the source of a small dipole field; the measured local field is the sum of the fields of all voxels. Iron and blood (paramagnetic) are positive, myelin and calcium (diamagnetic) negative. The map is relative: only differences between tissues are meaningful.

Limitations.

- **The inversion is ill-posed.** Some spatial patterns of χ make no field at all, so the conversion needs a constraint. TV (ADMM) assumes regions of constant χ and gives clean maps with sharp edges, but it flattens fine texture and lowers small, strong sources; TKD is quick but shows streaks and underestimates χ; closed-form L2 smooths; LSQR follows the data closely and keeps more noise and streaks.
- **Background removal** decides what counts as local. LBV keeps the tissue up to the mask's edge; V-SHARP is accurate inside but removes a rim at the edge; PDF can leave some background field near the edge. Errors there show as a smooth offset or shading in χ.
- **The mask** decides the reference (the mean over the mask) and the edge: a mask that includes noise or misses tissue changes all values.
- **Direction.** White matter's χ depends on its orientation to the main field; the map gives one value per voxel.
- **Fat** has a chemical shift that the field map takes for susceptibility. With water & fat ticked as well, the fat's phase is removed with the fat fraction of the water & fat fit before the field map is made, which needs echo times that suit water & fat (section 8); without it, fat in the body shows as false χ.
- **Short T_2*** (dense iron, near air or bone) leaves little signal at the later echoes; the field there is noisy and χ unreliable.
- **Flow** in vessels and **motion** add phase that is not susceptibility.
- QSM needs a 3D scan with near-isotropic voxels (section 8).

T_1, inversion recovery. Each pixel is fitted to

> $Signal = | A - B\ exp(-TI / T_1^*) |,\ T_1 = (B / A - 1)\ T_1^*$

T_1, saturation recovery. The signal as a function of TR is fitted to

> $Signal = A\ (1 - exp(-TR / T_1))$

T_1, variable flip angle. The spoiled gradient echo signal over the flip angles is fitted per pixel with DESPOT1; a series acquired as a CINE is fitted per cardiac frame. When the exam holds a flip-angle map scan (an AFI scan) that covers the scan (Step 1), each pixel is fitted with the flip angles it really got: the scan's flip angles times what the map measured there. The message box says how far the flip angle differed from the scan's over the mask; pixels that got less than half of it (the edges of a slab) are left out and get no T_1, and the message box says how many. When the map covers too little of the mask, or its values are not plausible, the message box says so and the fit uses the scan's flip angles everywhere. The pixel's plots then show the fit at the pixel's own flip angles, so that they agree with the map.

ORACLE. A balanced SSFP scan repeated with N RF phase increments gives every voxel a profile over the phase increments, whose shape depends on T_1, T_2, the flip angle and the off-resonance, and whose size on M_0. MAPapp calculates T_1, T_2 and the off-resonance from the profile directly, without fitting a curve by trial; M_0 is the scale that matches the predicted profile to the measured one. R^2 compares the profile the four results predict with the measured profile, so min R-square works as for the other scans. The off-resonance is given in Hz, between −1/(2 TR) and +1/(2 TR).

Model-based fits. The maps are estimated from the k-space itself: BART looks for the maps whose predicted k-space matches the measured one.

8. Scan protocols for water & fat and QSM

Water & fat and QSM are calculated from multi-gradient echo scans. MAPapp offers them only when the scan allows it (Step 5). The tables below give starting points for mice at 3 T, 7 T and 9.4 T; adapt the field of view and resolution to the animal and the organ.

Water & fat

What matters most:

- **No fat suppression.** With the fat saturated there is no fat signal to separate.
- **Echo spacing.** Fat and water are back in phase after the *in-phase period*: 2.30 ms at 3 T, 0.99 ms at 7 T, 0.73 ms at 9.4 T. An echo spacing of a whole number of in-phase periods (or half of one) sees fat the same way at every echo, and the scan cannot be separated (MAPapp then switches **water & fat** off and says why); close to it, the separation is poor (MAPapp warns). The echo spacings in the tables lie about a third or two thirds of a period past a whole number of periods, which separates water and fat best. They are short in absolute terms but reachable with ordinary gradients; set them to within about 0.05 ms (at 9.4 T the whole period is only 0.73 ms).
- **Bandwidth.** Fat is shifted against water along the readout by (fat frequency) / (bandwidth per pixel) pixels; the fat frequency is 430 Hz at 3 T, 1010 Hz at 7 T and 1360 Hz at 9.4 T. Use the highest bandwidth the echo spacing allows: with the values in the tables the shift stays at about one to two pixels.
- **Flip angle.** Fat recovers faster than water (shorter T_1); a small flip angle (3D) or a long TR (2D) keeps the fat fraction from being overestimated.
- **Readout.** Monopolar (fly-back) readouts are best; bipolar readouts are corrected by MAPapp.

2D multi-slice multi-gradient echo

	3 T	7 T	9.4 T
First echo time	2.0–2.5 ms	2.0–2.5 ms	2.0–2.5 ms
Echo spacing	3.05 ms (or 3.8 ms)	2.65 ms (or 3.3 ms)	2.45 ms (or 3.2 ms)
Echoes	8 (last echo about 24 ms)	8 (about 21 ms)	8 (about 19 ms)
Bandwidth per pixel	≥ 300 Hz	≥ 500 Hz	≥ 600 Hz
TR	400–800 ms	400–800 ms	400–800 ms
Flip angle	20–30°	20–30°	20–30°
Resolution (mouse)	0.15–0.2 mm in plane, slices 0.5–1 mm	0.1–0.15 mm, slices 0.5–1 mm	0.1 mm, slices 0.5 mm
Averages	1–2	1–2	1–2
Fat suppression	off	off	off

A shorter TR (100–200 ms) gives more slices per minute; use a smaller flip angle with it (15–20°), so that the faster recovery of fat raises the fat fraction less.

3D multi-gradient echo

	3 T	7 T	9.4 T
First echo time	2.0–2.5 ms	2.0–2.5 ms	2.0–2.5 ms
Echo spacing	3.05 ms (or 3.8 ms)	2.65 ms (or 3.3 ms)	2.45 ms (or 3.2 ms)
Echoes	8 (last echo about 24 ms)	8 (about 21 ms)	8 (about 19 ms)
Bandwidth per pixel	≥ 300 Hz	≥ 500 Hz	≥ 600 Hz
TR	15–30 ms	15–30 ms	15–30 ms
Flip angle	3–6°	3–6°	3–6°
Resolution (mouse, isotropic)	0.25–0.3 mm	0.15–0.2 mm	0.12–0.15 mm
Averages	1	1	1
Fat suppression	off	off	off

Avoid the in-phase echo spacings: 2.30 or 4.60 ms at 3 T, multiples of 0.99 ms at 7 T (2.0, 3.0, 4.0 ms) and of 0.73 ms at 9.4 T (2.2, 2.9, 3.7 ms). Both readout types work at these spacings; the longer spacing in brackets leaves more time for a monopolar readout or a lower bandwidth.

QSM

QSM needs a **3D** multi-gradient echo scan: the susceptibility is calculated from the field in all three directions, which 2D slices do not give, so MAPapp does not offer QSM for 2D scans.

What matters most:

- **Isotropic voxels** and a field of view that covers the whole organ.
- **Echo times** up to about the T_2^* of the tissue: the phase is measured best there.
- **One average.** The scanner sums averages, and averages acquired at slightly different frequencies do not add up well; MAPapp checks and corrects this, but one average is safer.
- **Fully sampled k-space,** or compressed sensing (**CS reco** first; it takes longer).
- **The scan's orientation** is read from the data; keep the animal still and the field of view as planned.
- In the body, fat disturbs the phase: choose an echo spacing from the water & fat tables, so that the same scan also gives water and fat.

3D multi-gradient echo for QSM (mouse brain)

	3 T	7 T	9.4 T
Resolution (isotropic)	0.2–0.25 mm	0.12–0.15 mm	0.1 mm
First echo time	2.5–3 ms	2.5 ms	2.5 ms
Echo spacing	3.5–4 ms	3 ms	2.5–3 ms
Echoes	8 (last echo 25–30 ms)	8 (last echo 20–25 ms)	8 (last echo 18–20 ms)
TR	last echo + 5–10 ms	last echo + 5–10 ms	last echo + 5–10 ms
Flip angle	10–15°	10–15°	10–15°
Readout	monopolar, flow compensated if available	monopolar	monopolar
Averages	1	1	1

9. CEST and MT scans

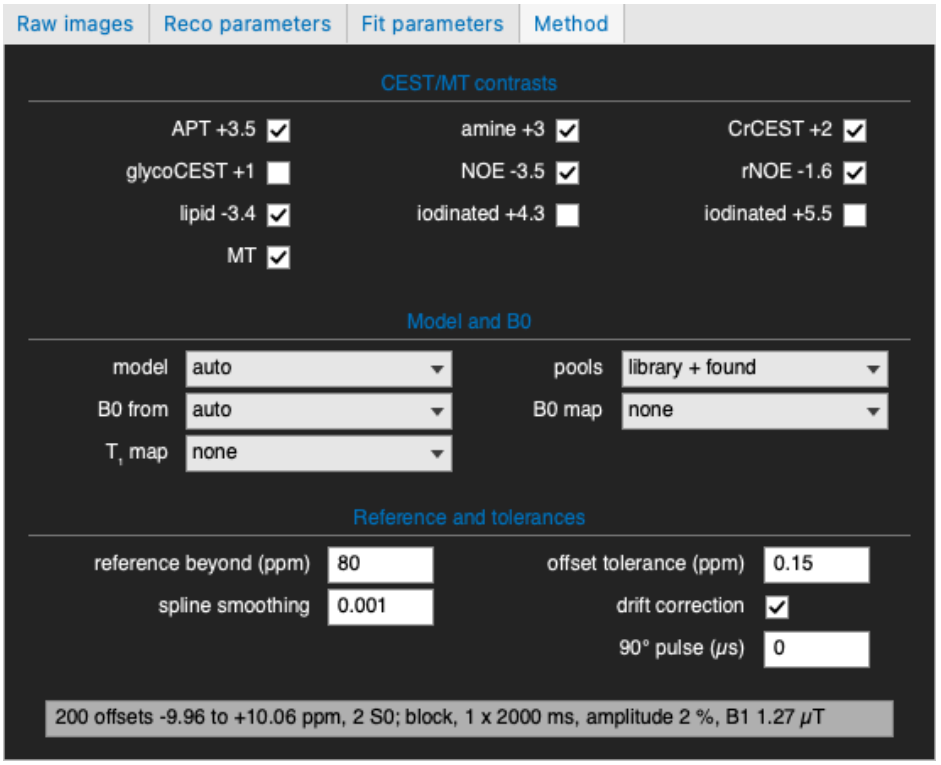
A CEST or MT scan repeats one image after a saturation at a series of frequency offsets from water, with one or more images without saturation (the reference, S_0). MAPapp recognizes such a scan when it is loaded: its points are the saturation offsets, shown in ppm under the point field (**S0** for a reference), and the line at the bottom of its Method tab says how many offsets the scan has, from where to where, how many references, and the saturation pulses as the scan describes them: their shape, number and duration, their amplitude in percent of the maximum and, where it is known, their B_1 in µT (see **90° pulse (µs)** below). An offset at or beyond the **reference beyond (ppm)** distance (80 ppm) counts as a reference. The point field can leave offsets out of the analysis; an offset the scan's parameters do not give is shown as **?** and left out, and so is a point the scan holds no data for (a scan stopped before its end), with a warning in the messages.

A sweep acquired without an unsaturated image takes its reference from the exam: the nearest scan before it and the nearest scan after it with the saturation off and the same protocol otherwise (sequence, matrix, slices, field of view, slice thickness, TR, TE, averages, orientation) are loaded with it, and the messages name them. Without such scans, the offsets farthest from water stand in for the reference, which the messages note: Z, MTR and the pool amplitudes are then relative to the signal there, which is a little lower than an unsaturated one; the asymmetry hardly changes. To use reference scans, acquire one with the saturation off before and after the sweep, with the same protocol. An EPI sweep takes no reference scans: its farthest offsets stand in.

A set of offsets may also be acquired one offset per scan: a scan of one saturated image at one offset (for example +3.5 ppm, then –3.5 ppm, then one far from water as the reference). Load any scan of the set: with **gather single-offset scans** ticked in the Reco parameters tab (the panel **CEST/MT scans**), MAPapp reads the exam's other scans of the same protocol (sequence, matrix, slices, field of view, slice thickness, TR, TE, averages, orientation and saturation pulses) with it, each with its own offset, together with an unsaturated scan of the protocol just before or after the set, and analyzes them as one Z-spectrum, sorted by offset. Each scan's offset is taken relative to the loaded scan's base frequency, so that a frequency set anew between the scans is accounted for. The messages list the scans used, their base frequencies and their offsets; a scan of another protocol in between (a localizer, a scan repeated with other settings) is passed over. The set ends where an offset repeats or a second unsaturated scan follows on the same side, so that a second set of the protocol in the exam (after an injection, say) stays a set of its own. The reference is an offset at or beyond the **reference beyond (ppm)** distance or an unsaturated scan. Untick the box before loading to analyze the scan alone. An EPI scan is analyzed alone.

A scan may hold a full sweep of offsets or only a few. MAPapp computes every map the offsets allow, for every contrast ticked in the Method tab, and leaves a map out where the offsets do not allow it. Within a map, a pixel whose offsets do not allow the value stays empty (black) rather than taking a value from offsets that do not belong to it; so does a pixel whose reference (S_0) image is 0.

The CEST/MT Method tab.



The CEST/MT Method tab after loading a CEST sweep

- **CEST/MT contrasts**: APT (+3.5 ppm), amine/GluCEST (+3.0 ppm), CrCEST (+2.0 ppm), glycoCEST/hydroxyl (+1.0 ppm), NOE (–3.5 ppm), rNOE (–1.6 ppm), lipid (–3.4 ppm, the fat line), the lines of iodinated contrast agents (**iodinated +4.3**: iopamidol +4.2 ppm, iohexol +4.3 ppm; **iodinated +5.5**: iopamidol) and MT (the offsets 5 ppm or more from water). Each ticked contrast gets every map that applies to it and a pool of its own in the Lorentzian fit. GlycoCEST and the iodinated lines start unticked. GlycoCEST lies so close to water that its pool can take up the shape of the water line rather than a hydroxyl exchange; tick it when the sample is expected to show glycogen or another hydroxyl signal. Tick the iodinated lines for a sample that holds such an agent to get their maps by name; unticked, the fit still finds such a line where the data show it (see **pools**). NOE maps near fat, or with incomplete fat suppression, include signal of the fat (the messages say so after a Lorentzian fit); the lipid maps, above all its Lorentzian difference (LD lipid), show the fat.
- **model**: what is fitted to the Z-spectra. **auto** takes the most the offsets allow: with a few offsets none (MTR and asymmetry at the measured offsets), with seven or more offsets within ±6 ppm, among them offsets on both sides of water within 1 ppm, a smoothing spline, with a full sweep (25 or more offsets within ±6 ppm, reaching past ±3.8 ppm on both sides) the Lorentzian pools. A model the offsets do not allow is replaced by the one they do, with a note in the messages.
- **pools**: the pools of the Lorentzian fit. **library + found** (the default): water, MT and a pool for each ticked contrast; where the fit then still leaves a dip in the spectrum that is more than noise, it adds a pool there, a *found* pool named by its position (for example **dip +5.5 ppm**), as long as that makes the fit clearly better. The pools are chosen once on the mean spectrum of the pixels analyzed, together with the pixels it fits worst, so that a line only some pixels hold (one tube of a phantom) is found as well; every pixel is then fitted with the same pools, and each pool has its maps. The messages say which pools were used and whether pools were found. **library only**: the ticked contrasts' pools alone; the messages name any dips they leave. The other choices are

fixed sets: water and MT; with amide and NOE; with amine; with hydroxyl. Their amine pool lies at +2 ppm and is named **Amine +2 ppm** (for example **Amplitude Amine +2 ppm**), apart from the amine contrast at +3.0 ppm. Two found pools whose positions round alike are told apart by a number (**dip +5.4 ppm** and **dip +5.4 ppm 2**).

- **B0 from**: where the shift of the water resonance comes from. **auto** takes the Z-spectrum where the offsets around water allow it, else a linked **B0 map**, else none; with none the maps are marked as B_0 uncorrected in the messages and the quality map.
- **B0 map** and **T_1 map**: maps of a scan fitted earlier in the same window, with the same matrix and slices: a T_1 map of a T_1 or ORACLE scan (for AREX), a B_0 map of an ORACLE scan (off-resonance), of a T_2* scan fitted with the B_0 correction, or the B_0 shift of a WASSR or CEST scan analyzed before (not the scan's own, when it is loaded again). To use them, load and fit that scan first, then load the CEST scan; the newest suitable map is chosen, **none** leaves it out.
- **reference beyond (ppm)**, **offset tolerance (ppm)** (how far a measured offset may lie from the offset a map asks for), **spline smoothing** and **drift correction** (with several references, the reference signal is followed through the scan, so that a slow signal drift does not show up in the maps).
- **gather single-offset scans**: read a scan of one saturation offset with the exam's other scans of its set (see above); it applies to the next load.
- **90° pulse (µs)**: the length of a rectangular pulse at 100 % amplitude that turns the magnetization by 90°, as found with an RF power calibration of the coil. Leave it at 0 to take the value from the scan's own RF calibration, for the scanner's standard 3D EPI, fast spin echo and inversion-recovery fast spin echo sequences. With either, the saturation's amplitude is also given as a B_1 in µT: in the line at the bottom of the tab, in the messages (which say where the value comes from, for example "saturation 2 % (B1 1.78 µT, from the scan's RF calibration ...)"), and in CEST_parameters.json. A value you enter is kept for the next start together with the RF calibration of the scan loaded at that moment, and is adjusted for the RF calibration of each later scan; it takes precedence over the scan's own value, which the messages then show beside it. For Gaussian saturation pulses the B_1 is their peak. Without either value, the amplitude is given in percent only.

The maps. Press **Fit CEST** after the reconstruction (and, where needed, the registration and the mask). The maps stand on four tabs; each tab's drop-down list above the map chooses which of its maps is shown, and **Auto** sets the scale from the shown map's values.

Tab	Maps	Needs
Z / MTR	MTR = 1 − Z at each measured offset (with more than 24 offsets: at the contrasts' offsets and the MT offsets); MTR of each contrast at its offset from water	any offsets
Asymmetry	MTRasym = Z(−Δω) − Z(+Δω) of each contrast downfield of water (APT, amine, CrCEST, glycoCEST, iodinated); its mean over the contrast's range of offsets; with 24 offsets or fewer, MTRasym at each measured pair ±Δω	both sides of the offset; the mean: seven or more offsets within ±6 ppm, some on each side of water within 1 ppm
Fitted pools	amplitude, width (ppm) and position (ppm from water) of each pool, named by the pool: **Amplitude MT**, **Width NOE**, **Amplitude APT**, **Position dip +5.5 ppm**; per contrast the Lorentzian difference (the water and MT lines minus the measured Z), MTRrex and, with a T_1 map, AREX (1/s)	a full sweep
B0 & quality	the B_0 shift of water (ppm); the B_0 shift the Lorentzian fit gives (**B0 Lorentzian**, ppm); the R^2 and the residual (its root mean square, in Z) of the Lorentzian fit; the number of offsets used; the quality flags	—

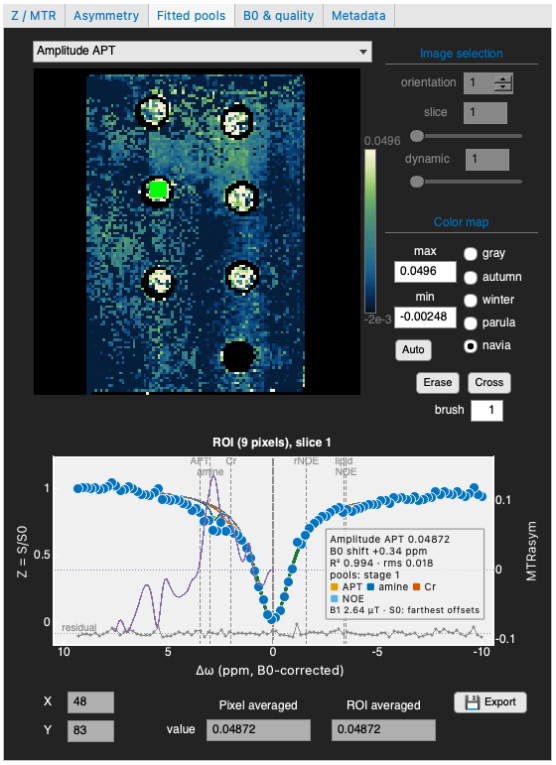
The quality flags add up: 1 B_0 uncorrected, 2 no B_0 shift found for the pixel, 4 a B_0 shift (from a B_0 map) larger than half the offset step, 8 a Lorentzian fit with R^2 below 0.95, 16 a missing value in the Z-spectrum.

Pixels and ROIs. A click on a map, or an ROI, shows the value of each tab's map there, and a plot on every tab, titled with the pixel or the ROI and the slice: the Z-spectrum, $Z = S/S_0$ against the offset from water (corrected for the B_0 shift where there is one). Offsets are plotted downfield (positive ppm) at the left, as is usual in NMR. The measured points are blue, points left out with the point field hollow gray. With a full sweep the Lorentzian fit is drawn as a line, its water and MT background dashed, and each pool as a colored area between the background and the fit, a found pool's edge dashed; under Z = 0 a narrow strip shows the residual, the measured Z minus the fit, which should scatter evenly around its dotted middle line. The pools are chosen for the pixel's or ROI's own spectrum, so an ROI in one tube may show a found pool the maps' pools do not have, or the other way round. With fewer offsets the spline is drawn, and with a few offsets the points alone. Dashed gray lines mark water (0 ppm) and each ticked contrast's offset, named at the top where a name has room (a name that would overlap another is left out, its line kept). The asymmetry MTRasym is drawn on the positive offsets as a thin line on its own scale at the right. A small box in the emptiest corner gives the value of the tab's map there, the B_0 shift, the fit's R^2 and residual (rms), whether pools were found ('pools: stage 1': the ticked contrasts' pools explain the spectrum; 'stage 2: 1 found': a pool was added), the pools drawn in their colors (■ a contrast's pool, ▲ a found pool), the saturation's B_1 and where the reference comes from.

Export on a CEST tab writes, in the scan's …`/Data/<exam>/MapAnalysis/<tag>CEST/<n>` folder (a new numbered folder each time; for a scan outside the scanner's folders, a folder you choose):

- `roiZspectrum-<tag>.csv`: one row per measured offset in increasing order: the offset from water (B_0-corrected) and as set, Z, and where they exist the fit, the background and each pool at that offset, over the whole sweep (or, without the fit, the spline), and MTRasym;

- `roiZspectrum-<tag>.png`: the plot as it is shown;
- `roiZspectrum-<tag>-figure.png`: the plot as a figure for a paper: 16 × 10 cm at 300 dpi, white with dark text, a legend of the measured points, the fit, the background, the contrasts' pools and the found pools, the asymmetry on the right axis, the residual strip, and a caption line with the scan, the ROI, the B_0 shift, R^2, the residual, whether pools were found, the B_1, the reference and the app's version;
- `roiZspectrum-<tag>.xlsx`, a workbook with four sheets: **Summary** (the scan, the ROI, B_0, R^2, the residual, the pools, the saturation and its B_1), **Pools** (each pool: kind, amplitude, width and position), **Spectrum** (each measured offset: Z, whether it was used, the fit, the background, the residual, each pool and MTRasym) and **Maps** (each map's mean, SD, median, smallest and largest value and number of pixels in the ROI).

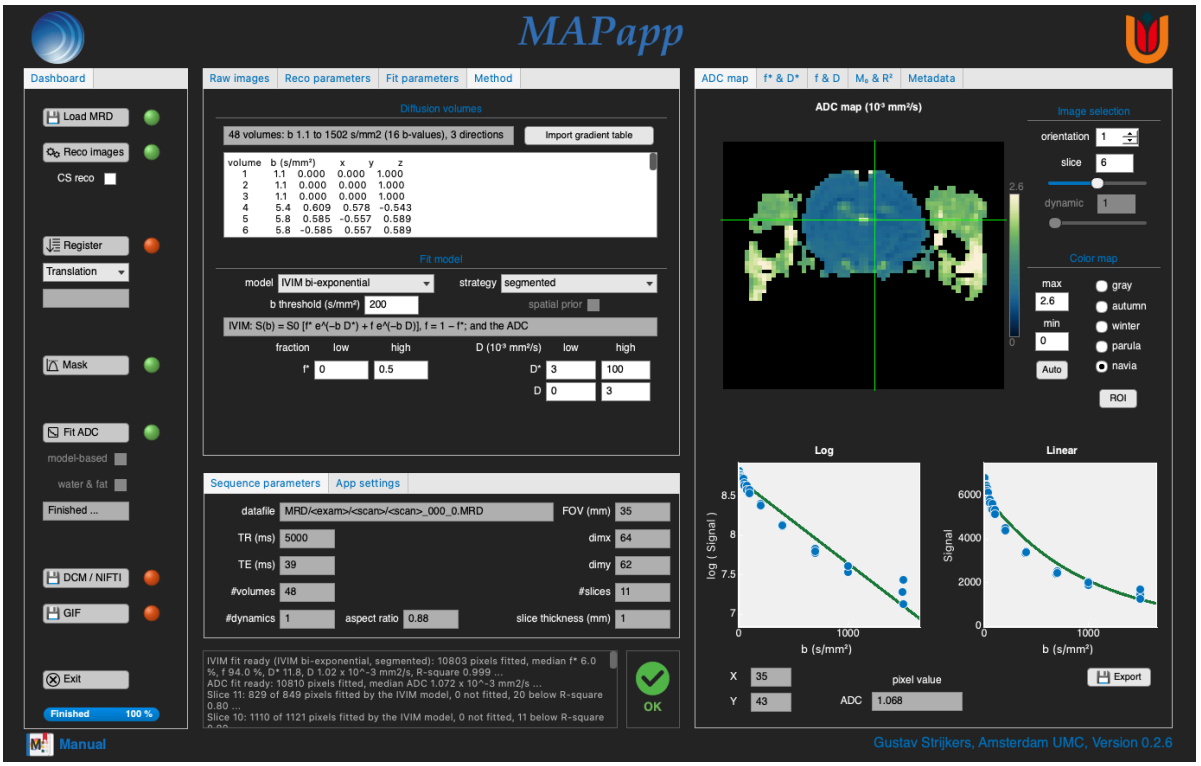


The Fitted pools tab of a CEST sweep of a tube phantom: the APT amplitude map with an ROI in one tube, and the ROI's Z-spectrum with its fitted pools

Exports. DCM / NIFTI writes every map as its own DICOM series and NIfTI file, with a file CEST_parameters.json beside them that lists the offsets, the saturation pulses with their amplitude and B_1 and where the B_1 comes from, the options and every map's unit; the export's report lists it with the other files. BIDS defines no CEST maps, so none are written. **GIF** writes an animation per map, and a picture of the window with each CEST tab that shows a map.

10. Diffusion scans

A diffusion-weighted EPI scan repeats one image with a diffusion weighting, volume after volume, each volume with its b-value and gradient direction. MAPapp recognizes such a scan when it is loaded, when its parameters give the b-value of every volume and at least two different b-values; a diffusion scan that cannot be mapped is refused with the reason in the messages. Its map is the apparent diffusion coefficient (ADC): the map tabs are **ADC map** and **M_0 & R^2**, the Fit button reads **Fit ADC**, and the Method tab shows the scan's volumes and the fit model. On request the fit adds the intravoxel incoherent motion (IVIM) model, which separates the diffusion of the water in the tissue (and, with the tri-exponential model, in a second, faster compartment such as CSF) from the pseudo-diffusion of the blood in the capillaries; its maps get tabs of their own.



The MAPapp window after an IVIM fit of a diffusion scan of a rat brain: the ADC map and the Method tab

The b-values. The protocol's b-value of a volume counts only its diffusion gradients. The spins also see the imaging gradients (slice selection, refocusing, readout), and these add a little weighting of their own and change that of the diffusion gradients in a way that depends on the direction. For the scanner's spin-echo diffusion EPI sequences, with an EPI readout or one echo per shot, MAPapp therefore works out every volume's b-matrix from all the gradients the sequence plays. Its b-value is the b-matrix's trace and its direction the b-matrix's main direction. The volumes without diffusion gradients get a small b-value (around 1 to 3 s/mm²), and the directions of one protocol b-value get b-values a little apart (a few tenths of a percent at high b-values, several percent at the lowest) and directions a few degrees from the protocol's at the lowest b-values. The messages say which b-values are used as the scan loads, and the fits use them volume by volume. This applies to the current diffusion sequences, their older versions (from 2020 on) and the lab's corrected version. For other sequences the protocol's b-values and directions are used. When the scan's protocol computed its b-values with another gradient strength than the one the sequence plays with, a warning gives both and the difference in percent; the b-values follow the gradient strength played.

Echo timing. With some protocols the sequence reads the center of k-space of the diffusion-weighted volumes about 1 ms earlier than that of the volumes without diffusion weighting. This gives their signal a step that does not depend on b, which an IVIM fit would take for perfusion. A warning says so as the scan loads and names the shortest echo time (TE) at which the same protocol avoids it. The IVIM fit then leaves the volumes without diffusion weighting out where enough low b-values remain, and says so; the ADC fit and the exports keep every volume of the point field. The Sequence parameters tab shows the scan's TE.

Loading from DICOM. A diffusion series can also be opened from the DICOM files the scanner wrote of it: choose any of its `.dcm` files (the series in that folder is loaded), or give the scan's folder. Only diffusion series are read from DICOM; another series is refused with a message naming what the scan is. The images are the scanner's own, so there is nothing to reconstruct: **Reco** only scales them (and suppresses ringing with **de-ring**); **de-noise** and compressed sensing do not apply to them. The b-value and direction of every volume come from the scan's protocol when the scan's MRD file lies where the scanner keeps it (the MRD folder of the same exam and scan number); a warning names the volumes whose DICOM files give other values, and the protocol's are used. Without the MRD file they come from the files, and a warning names any volume whose signal contradicts its b-value (a volume with a large b-value that is as bright as the volumes without diffusion weighting): check the b-values, and give the right ones with **Import gradient table** on the Method tab. The exports of a series loaded from DICOM go to the exam's folder beside its DICOM folder, named by the scan number.

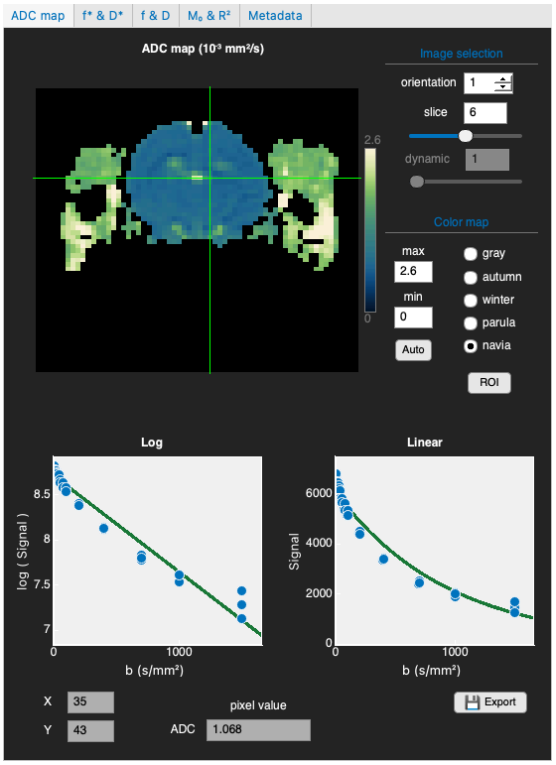
Loading. The **de-noise** box of the Reco parameters tab acts as a diffusion scan is loaded from its MRD file: set it before loading. After the load the box shows whether the scan was denoised; to change that, set the box and load the scan again (a box changed later is put back at the next reconstruction, with a message).

The volumes. The points of a diffusion scan are its volumes: the line under the point field shows their b-values in s/mm², in the order of the field's digits, and the point field leaves volumes out of the fit (a 0 for each volume left out). The volumes kept must hold at least two different b-values, at least 100 s/mm² apart; otherwise the fit is refused with a message and the maps stay as they were.

Registration and mask. Register moves every volume onto the volume of the lowest b-value (the first b = 0 volume), which has the most signal. **Mask** makes the mask from that volume as for the other scans, and a typed threshold applies to it.

The fit. Every masked pixel is fitted with $S(b) = M_0 \exp(-b \cdot ADC)$, a straight line through the logarithm of the signal against b over the volumes kept. The ADC is in 10^{-3} mm²/s (brain tissue lies at about 0.5 to 1.2, water at body temperature near 3). R^2 says how straight that line is over the b-values: with only two different b-values the line runs through both and R^2 is 1, which the messages say. A pixel whose R^2 is below the **min R-square** field loses its ADC and M_0 and keeps its R^2, so that the pixels left out can be seen on the R^2 map.

The ADC map tab. The map in 10^{-3} mm²/s on its scale; **Auto** sets the scale from 0 to twice the mean ADC of the fitted pixels. A click on a map gives the pixel's ADC; an ROI gives the mean ADC of its fitted pixels in **Pixel averaged** and the ADC of a fit to the ROI's mean signal in **ROI averaged**. The two plots show the signal of the pixel or ROI against b, linear and logarithmic, with the fitted decay; a volume left out of the fit is red.



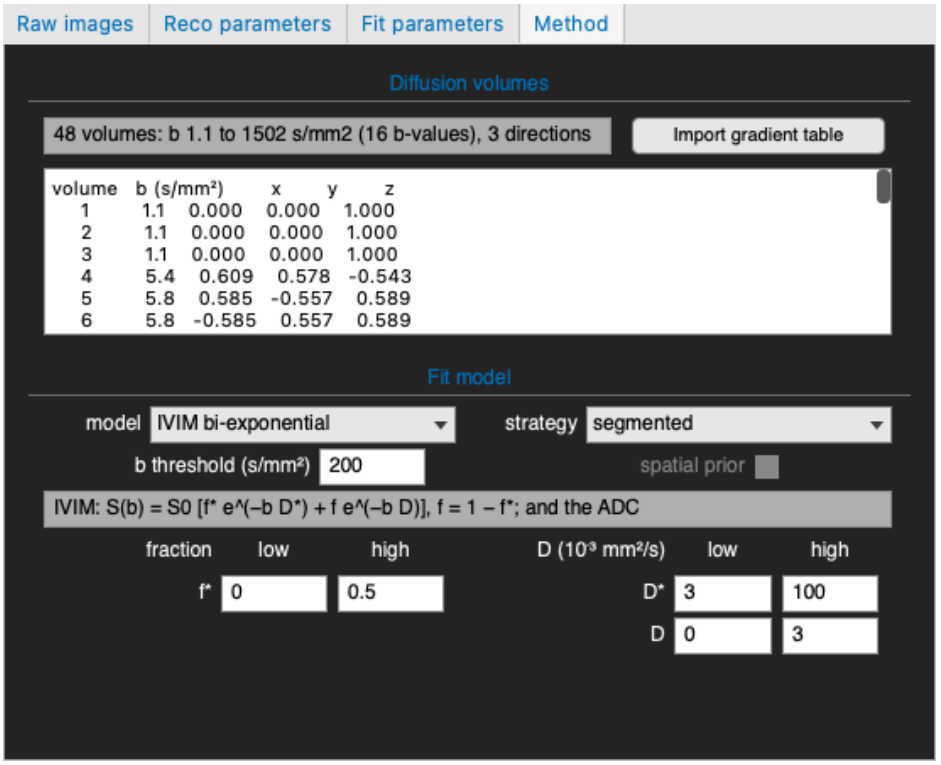
The ADC map tab with the decay of one pixel

The Method tab. Diffusion volumes: a line with the number of volumes, the range of b-values and the number of directions, and a table with one line per volume: its number, its b-value (s/mm²) and its gradient direction (x, y, z; zero for b = 0). **Import gradient table** replaces the b-values and directions of the loaded scan by those of a table file: a text file with one line per volume, in the order of the volumes, each with the b-value (s/mm²) and the three components of the direction (0 0 0 for b = 0), separated by spaces, tabs, commas or semicolons; a line starting with % or # is a comment. A table without one line per volume, with a negative b-value or with a direction component beyond −1 … 1 is refused with a message and changes nothing. A table taken shows at once in the volume table and under the point field; maps fitted before are withdrawn (the export buttons go off) until **Fit ADC** fits them again with the table's b-values. The dialog opens in the folder of the last table. **Fit model**: the model the fit uses, chosen with these controls (only those of the chosen model and strategy can be changed; the app keeps every choice for the next start):

- **model**: **ADC (mono-exponential)** fits the ADC alone. The IVIM models have one perfusion compartment, the blood in the capillaries, with its pseudo-diffusion coefficient D* and signal fraction f*, and one or two diffusion compartments. **IVIM bi-exponential** fits the ADC and $S(b) = S_0 [f^* \exp(-b D^*) + f \exp(-b D)]$, $f = 1 - f^*$, with the diffusion compartment's (the tissue's) diffusion coefficient D and fraction f. **IVIM tri-exponential** fits the ADC and $S(b) = S_0 [f^* \exp(-b D^*) + f_1 \exp(-b D_1) + f_2 \exp(-b D_2)]$, $f_2 = 1 - f^* - f_1$, with a slower diffusion compartment (D_1, f_1; brain tissue, for example) and a faster one (D_2, f_2; free water such as CSF, for example). The line under the controls shows the chosen model's equation.
- **strategy** (IVIM only): **segmented** fits the diffusion compartments from the b-values at or above the threshold first, then f* and D* with them held; **full NLLS** fits every parameter at once, starting from the segmented result; **spatially regularized (IDEAL)** fits each slice from a coarse version (blocks of 8 × 8 pixels) up to full resolution, each pixel kept near its surroundings' result, which

gives smoother maps; **Bayesian** pulls a pixel whose data say little towards the values of the whole slice.

- **b threshold (s/mm²)** (IVIM only): the b-values at or above it give the diffusion compartments (200 by default; 300 is common in the brain). The fit needs at least two different b-values below it, and two (bi) or four (tri) at or above it.
- **spatial prior** (Bayesian only): also pulls each pixel towards its neighbors.
- **bounds**, a lower and an upper value per parameter of the chosen model; only the chosen model's are shown: the fractions (**f***, **f_1**) as fractions (0.5 is 50 %; f and f_2 are what the others leave of 1), the diffusivities (**D***, **D**, **D_1**, **D_2**) in 10^{-3} mm²/s. A fitted value stays within its bounds. By default, bi-exponential: f* 0 to 0.5, D* 3 to 100, D 0 to 3; tri-exponential: f* 0 to 0.5, f_1 0 to 0.99, D* 5 to 100, D_1 0 to 1.5, D_2 1.5 to 4. The tri-exponential bounds keep the three diffusivities apart (D_1 below D_2 below D*), so that the fit cannot swap two compartments. For the kidney raise the upper bound of f* to 0.7; for fast flow that of D* to 500.



The Method tab of a diffusion scan: its volumes and the fit model

The IVIM fit. With an IVIM model, **Fit ADC** fits the ADC first, as with the ADC alone (the ADC map always shows the mono-exponential ADC), then the IVIM model. The volumes of each b-value are averaged over their directions first, so that the IVIM fit sees one signal per b-value, as the ADC does. A pixel whose IVIM fit has an R^2 below the **min R-square** field keeps its R^2 and loses its IVIM values. The messages give the number of pixels fitted and the median of each map.

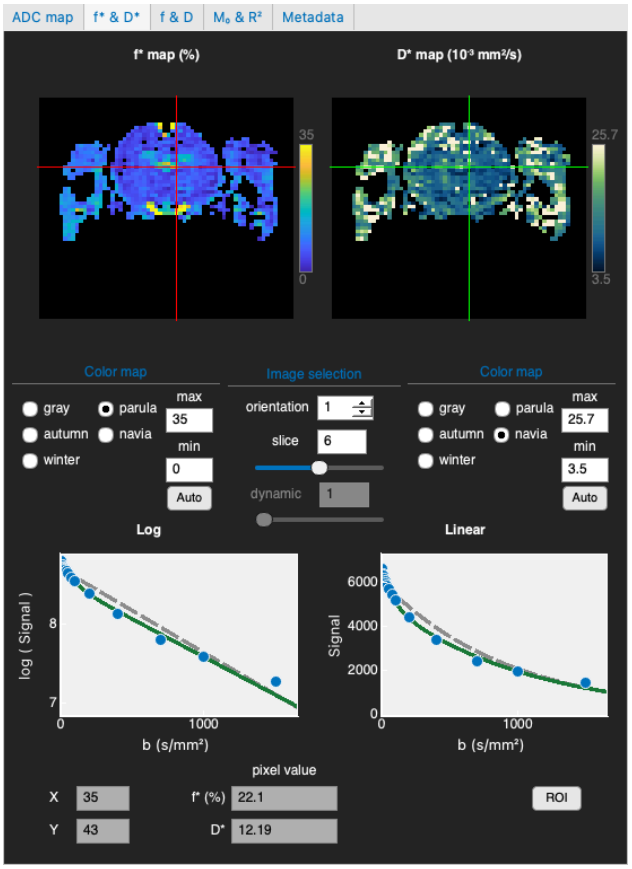
The IVIM tabs. After an IVIM fit these tabs show between **ADC map** and **M_0 & R^2**:

- **f* & D***: the perfusion compartment, its fraction f* in percent beside its pseudo-diffusion coefficient D* in 10^{-3} mm²/s.
- **f & D** (bi-exponential) or **f_1 & D_1** (tri-exponential): the (slower) diffusion compartment, its fraction in percent beside its diffusion coefficient in 10^{-3} mm²/s.

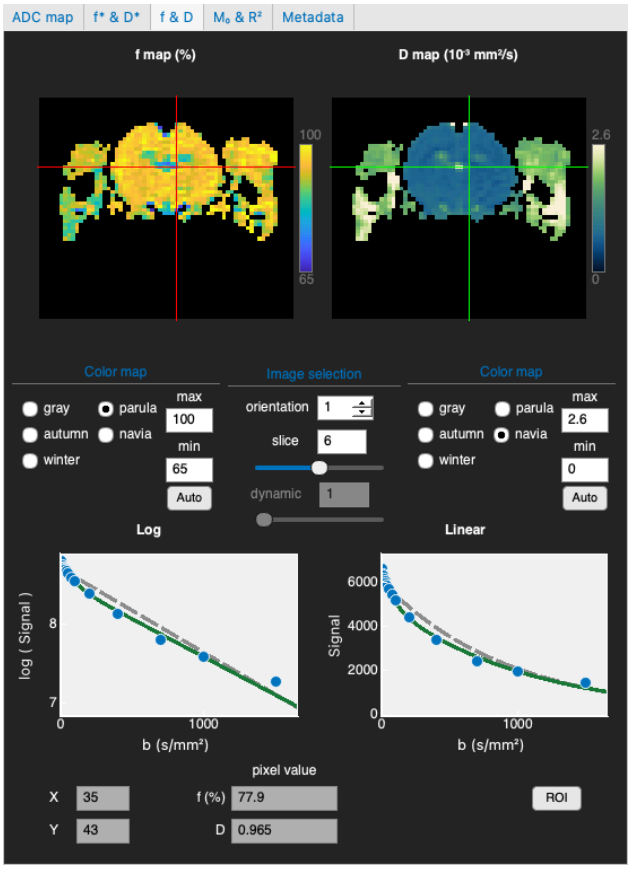
- **f_2 & D_2** (tri-exponential only): the faster diffusion compartment, in the same way.

Each map has its own color bar, scale and color map; every diffusivity is shown in the ADC map's color map. **Auto** sets a fraction's scale from the value 1 % of the fitted pixels stay below to the value 99 % stay below, in whole percent (so f* runs from about 0 and f up to 100), a diffusion coefficient's (D, D_1, D_2) by the ADC map's rule, so that they can be compared with the ADC, and D*'s from about its lowest value to the value 90 % of its pixels stay below, counting only the pixels whose value lies inside the limits set for it on the Method tab (a pixel that ends on a limit says little about D*), as its values spread over more than a factor ten. The **M_0 & R^2** tab shows the IVIM fit's S_0 and R^2; its headings say so. After a fit of the ADC alone the IVIM tabs go and that tab shows the ADC fit's M_0 and R^2 again. A click on a map gives the pixel's values on every tab, and each IVIM tab's two plots (linear and logarithmic) show the pixel's or ROI's signal per b-value (the averages the fit used), the fitted IVIM curve and, dashed, the ADC's straight decay for comparison. For an ROI, **ROI averaged** holds the values of a fit to the ROI's mean signal.

f* & D*



f & D



The command line. `--process <scan> --fit` fits the ADC; add `--ivim bi` or `--ivim tri` for the IVIM fit, and `--ivim-method segmented`, `nlls`, `ideal` or `bayes` for its strategy (segmented by default). It prints the pixels fitted and the medians.

Exports. The exports of a diffusion scan go into folders named `<tag>ADC` (step 10):

- **DCM / NIFTI** writes the maps as DICOM series and NIfTI files: **ADC** (10^{-3} mm²/s), **M0** (after an IVIM fit the IVIM fit's S_0, as the M_0 & R^2 tab shows) and **R2**, and after an IVIM fit **fstar**, **Dstar**, **f** and **D** (bi-exponential) or **fstar**, **Dstar**, **f1**, **D1**, **f2** and **D2** (tri-exponential), the fractions in percent

and the diffusivities in 10^{-3} mm²/s. A reader that applies the DICOM rescale slope (as NIfTI readers do with the header's scaling) gets these units back. The ADC map also goes into the BIDS folder, under `derivatives/mapapp/sub-<tag>/dwi`, as `_ADCmap` in mm²/s.

- Beside the maps it writes the diffusion-weighted images the maps were fitted from as the series **DWI**: in DICOM with the b-value and gradient direction of every volume, and in NIfTI as `DWI.nii` with `DWI.bval` and `DWI.bvec`, the gradient directions in the sense the images agree with. After a **Rigid** or **Affine** registration the directions in `DWI.bvec` are turned with the images they belong to; the DICOM files keep the directions as measured. Where the b-values were worked out from the sequence's gradients, `DWI_bmatrix.txt` lies beside them: the full b-matrix of every volume, one volume per line, Bxx Bxy Bxz Byy Byz Bzz in s/mm², in the axes of `DWI.bvec` and turned as it is. The `.bval` and `.bvec` then hold the b-matrices' traces and main directions, the volumes without diffusion weighting included: their imaging gradients give them a few s/mm² along a direction of their own, which a tensor fit from the `.bval` and `.bvec` then takes into account. Where the directions allow a diffusion tensor (six or more directions and a b = 0 volume) scripts for MRtrix3 and DSI Studio lie beside them (`tractography_MRtrix_<tag>.sh` and `.bat`, `tractography_DSIStudio_<tag>.sh` and `.bat`; written, not run), which make the diffusion maps and the tractography. The series also goes into the BIDS folder as the dwi run of the exam, `sub-<exam>/ses-1/dwi`, with its `.bval`, `.bvec` and sidecar, the image axes named in the animal's frame. `recoparameters_<tag>.txt`, in the DICOM and the NIfTI folder, says how the series was made: the scan, its geometry, the reconstruction and the diffusion table.
- **GIF** writes the ADC, M_0 and R^2 maps, and after an IVIM fit its maps, as animations in the scales and color maps shown, with pictures of the window's M_0 & R^2, ADC map and IVIM tabs.
- The **Export** button of the ADC map and D tabs writes the b-values and the signal of the pixel or ROI, `roiData-<tag>.csv` (one line per volume, without a header), and beside it `roiFit-<tag>.csv` with a header line and per volume the b-value, the signal, whether the fit used the volume (1 or 0), the ADC fit's curve and, after an IVIM fit, the IVIM fit's curve at that b-value.
- The report of each export holds the volumes with their b-values and directions, the fit model, the IVIM strategy, the b-values used and the median of each map. Where the b-values were worked out from the sequence's gradients, it also holds the protocol's b-values and every volume's b-matrix.

11. Credits

- **Diffusion**: E.O. Stejskal, J.E. Tanner, Spin diffusion measurements: spin echoes in the presence of a time-dependent field gradient, J Chem Phys 42(1), 288–292 (1965).
- **APT and the asymmetry**: J. Zhou, J.-F. Payen, D.A. Wilson, R.J. Traystman, P.C.M. van Zijl, Using the amide proton signals of intracellular proteins and peptides to detect pH effects in MRI, Nat Med 9(8), 1085–1090 (2003).
- **B_0 from the water line (WASSR)**: M. Kim, J. Gillen, B.A. Landman, J. Zhou, P.C.M. van Zijl, Water saturation shift referencing (WASSR) for chemical exchange saturation transfer (CEST) experiments, Magn Reson Med 61(6), 1441–1450 (2009).
- **Lorentzian fit of Z-spectra**: M. Zaiss, B. Schmitt, P. Bachert, Quantitative separation of CEST effect from magnetization transfer and spillover effects by Lorentzian-line-fit analysis of z-spectra, J Magn Reson 211(2), 149–155 (2011).
- **MTRrex and AREX**: M. Zaiss, J. Xu, S. Goerke, I.S. Khan, R.J. Singer, J.C. Gore, D.F. Gochberg, P. Bachert, Inverse Z-spectrum analysis for spillover-, MT-, and T1-corrected steady-state pulsed CEST-MRI – application to pH-weighted MRI of acute stroke, NMR Biomed 27(3), 240–252 (2014).
- **Iodinated contrast agents' CEST lines**: S. Aime, D. Calabi, L. Biondi, M. De Miranda, S. Ghelli, L. Paleari, C. Rebaudengo, E. Terreno, Iopamidol: exploring the potential use of a well-established x-ray contrast agent for MRI, Magn Reson Med 53(4), 830–834 (2005).
- **Bayesian information criterion**: G. Schwarz, Estimating the dimension of a model, Ann Stat 6(2), 461–464 (1978).
- **ORACLE**: N.M.J. Plähn, Y. Safarkhanlo, B.C. Açikgöz, A.L.C. Mackowiak, P. Radojewski, G. Bonanno, E.S. Peper, R. Wiest, J.A.M. Bastiaansen, ORACLE: an analytical approach for T1, T2, proton density, and off-resonance mapping with phase-cycled balanced steady-state free precession, Magn Reson Med 93(4), 1657–1673 (2025).
- **BART toolbox**: mrirecon.github.io/bart. M. Uecker et al., Berkeley Advanced Reconstruction Toolbox, Proc. Intl. Soc. Mag. Reson. Med. 23, 2486 (2015).
- **sigpy**: Frank Ong and Michael Lustig, SigPy: a python package for high performance iterative reconstruction, Proc. Intl. Soc. Mag. Reson. Med. 27, 4819 (2019).
- **elastix**: S. Klein, M. Staring, K. Murphy, M.A. Viergever, J.P.W. Pluim, elastix: a toolbox for intensity based medical image registration, IEEE Transactions on Medical Imaging 29(1), 196–205 (2010). elastix.dev
- **PCA denoising**: J. Veraart, D. S. Novikov, D. Christiaens, B. Ades-Aron, J. Sijbers, E. Fieremans, Denoising of diffusion MRI using random matrix theory, NeuroImage 142, 394–406 (2016).
- **Gibbs ringing suppression**: E. Kellner, B. Dhital, V.G. Kiselev, M. Reisert, Gibbs-ringing artifact removal based on local subvoxel-shifts, Magn Reson Med 76(5), 1574–1581 (2016).
- **Water & fat separation**: the ISMRM Fat-Water Toolbox: H.H. Hu, P. Börnert, D. Hernando et al., ISMRM workshop on fat-water separation: insights, applications and progress in MRI, Magn Reson Med 68(2), 378–388 (2012); its fw3pluspoint method: S.D. Sharma, H.H. Hu, K.S. Nayak,

Accelerated water-fat imaging using restricted subspace field map estimation and compressed sensing, Magn Reson Med 67(3), 650–659 (2012).

- **DESPOT1**: S.C.L. Deoni, B.K. Rutt, T.M. Peters, Rapid combined T1 and T2 mapping using gradient recalled acquisition in the steady state, Magn Reson Med 49(3), 515–526 (2003).
- **Water & fat separation methods**: graph cut: D. Hernando, P. Kellman, J.P. Haldar, Z.-P. Liang, Robust water/fat separation in the presence of large field inhomogeneities using a graph cut algorithm, Magn Reson Med 63(1), 79–90 (2010), with V. Kolmogorov, R. Zabih, What energy functions can be minimized via graph cuts?, IEEE Trans Pattern Anal Mach Intell 26(2), 147–159 (2004); hierarchical IDEAL: J. Tsao, Y. Jiang, Hierarchical IDEAL: fast, robust, and multiresolution separation of multiple chemical species from multiple echo times, Magn Reson Med 70(1), 155–159 (2013); region growing: H. Yu, S.B. Reeder, A. Shimakawa, J.H. Brittain, N.J. Pelc, Field map estimation with a region growing scheme for iterative 3-point water-fat decomposition, Magn Reson Med 54(4), 1032–1039 (2005); IDEAL: S.B. Reeder, A.R. Pineda, Z. Wen et al., Iterative decomposition of water and fat with echo asymmetry and least-squares estimation (IDEAL), Magn Reson Med 51(1), 35–45 (2004); H. Yu, C.A. McKenzie, A. Shimakawa et al., Multiecho reconstruction for simultaneous water-fat decomposition and T2* estimation, J Magn Reson Imaging 26(4), 1153–1161 (2007); variable projection: G.H. Golub, V. Pereyra, The differentiation of pseudo-inverses and nonlinear least squares problems whose variables separate, SIAM J Numer Anal 10(2), 413–432 (1973); D. Hernando, J.P. Haldar, B.P. Sutton, J. Ma, P. Kellman, Z.-P. Liang, Joint estimation of water/fat images and field inhomogeneity map, Magn Reson Med 59(3), 571–580 (2008); the fat fraction: H. Yu, A. Shimakawa, C.D.G. Hines et al., Combination of complex-based and magnitude-based multiecho water-fat separation for accurate quantification of fat-fraction, Magn Reson Med 66(1), 199–206 (2011); D. Hernando, C.D.G. Hines, H. Yu, S.B. Reeder, Addressing phase errors in fat-water imaging using a mixed magnitude/complex fitting method, Magn Reson Med 67(3), 638–644 (2012); the bipolar readout: W. Lu, H. Yu, A. Shimakawa, M. Alley, S.B. Reeder, B.A. Hargreaves, Water-fat separation with bipolar multiecho sequences, Magn Reson Med 60(1), 198–209 (2008); coil combination: D.O. Walsh, A.F. Gmitro, M.W. Marcellin, Adaptive reconstruction of phased array MR imagery, Magn Reson Med 43(5), 682–690 (2000).
- **Susceptibility mapping (QSM)**: the dipole field: R. Salomir, B.D. de Senneville, C.T.W. Moonen, A fast calculation method for magnetic field inhomogeneity due to an arbitrary distribution of bulk susceptibility, Concepts Magn Reson B 19(1), 26–34 (2003); J.P. Marques, R. Bowtell, Application of a Fourier-based method for rapid calculation of field inhomogeneity due to spatial variation of magnetic susceptibility, Concepts Magn Reson B 25(1), 65–78 (2005). Phase unwrapping: W. Xu, I. Cumming, A region-growing algorithm for InSAR phase unwrapping, IEEE Trans Geosci Remote Sens 37(1), 124–134 (1999). Background field removal: LBV, D. Zhou, T. Liu, P. Spincemaille, Y. Wang, Background field removal by solving the Laplacian boundary value problem, NMR Biomed 27(3), 312–319 (2014); V-SHARP, F. Schweser, A. Deistung, B.W. Lehr, J.R. Reichenbach, Quantitative imaging of intrinsic magnetic tissue properties using MRI signal phase, NeuroImage 54(4), 2789–2807 (2011), and B. Wu, W. Li, A. Guidon, C. Liu, Whole brain susceptibility mapping using compressed sensing, Magn Reson Med 67(1), 137–147 (2012); PDF, T. Liu, I. Khalidov, L. de Rochefort et al., A novel background field removal method for MRI using projection onto dipole fields (PDF), NMR Biomed 24(9), 1129–1136 (2011). Dipole inversion: TKD, K. Shmueli, J.A. de Zwart, P. van Gelderen et al., Magnetic susceptibility mapping of brain tissue in vivo using MRI phase data, Magn Reson Med 62(6), 1510–1522 (2009); LSQR, C.C. Paige, M.A. Saunders,

LSQR: an algorithm for sparse linear equations and sparse least squares, ACM Trans Math Softw 8(1), 43–71 (1982); TV and closed-form L2, B. Bilgic, A.P. Fan, J.R. Polimeni et al., Fast quantitative susceptibility mapping with L1-regularization and automatic parameter selection, Magn Reson Med 72(5), 1444–1459 (2014); C. Milovic, B. Bilgic, B. Zhao, J. Acosta-Cabronero, C. Tejos, Fast nonlinear susceptibility inversion with variational regularization, Magn Reson Med 80(2), 814–821 (2018).

- **Otsu's threshold**: N. Otsu, A threshold selection method from gray-level histograms, IEEE Trans Syst Man Cybern 9(1), 62–66 (1979).
- **Color maps**: F. Crameri, Scientific colour maps (navia, lipari), Zenodo, doi:10.5281/zenodo.1243862.
- **NIfTI conversion**: dcm2niix: C. Rorden et al., github.com/rordenlab/dcm2niix.

The standard brains MAPapp finds the brain with are made, changed, from:

- **Mouse:** the in-vivo C57BL/6J mouse brain atlas of Brookhaven National Laboratory and the National High Magnetic Field Laboratory (Ma et al. 2005, PMID 16165303; Ma et al. 2008, PMID 18958199; supported by NIH grants R01 EB 00233-04 and P41 RR16105 and the National High Magnetic Field Laboratory), nitrc.org/projects/c57bl_mr_atlas, licensed CC BY-NC-SA 4.0; the mouse template in MAPapp is shared under the same license.
- **Rat:** the SIGMA rat brain templates and atlases, version 2.0 (Barrière et al., Nature Communications 10, 5699, 2019), zenodo.org/records/10635831, licensed CC BY 4.0; MAPapp's rat template is changed from it (resampled and cropped).